



WHO INFLUENZA CENTRE
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Report

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Influenza activity, February to September 2009

Influenza during this period was dominated by the emergence of the pandemic A(H1N1) virus. Although infections in many countries continued to be due largely to imported cases (travellers), the virus has predominated in most of the countries where it has circulated more widely. Seasonal influenza A(H1N1), A(H3N2) and B viruses have continued to circulate at low levels. Thirty six cases of H5N1 infection were reported since the beginning of February, including four fatal cases in Vietnam during February to April and 32 cases in Egypt, four of which were fatal.

During February and March in the northern hemisphere, influenza continued to be relatively mild. H3N2 viruses predominated in Europe, where the season was effectively over by mid April (week 16), although influenza B rose to prominence late in the season, accounting for 16% of viruses in total, the majority (97%) of which were of the B/Victoria/2/87 lineage (Figure 1). In contrast, circulation of H3N2 viruses was relatively low in North America where H1N1 (seasonal) predominated in the USA and influenza B was prevalent together with H1N1 viruses in Canada, as they were in Japan. H1N1, H3N2 and B viruses co-circulated in many northern hemisphere and tropical countries in Africa and Asia. Increased activity in some Asian countries from June to August included outbreaks of H3N2 in China.

Following detection of the pandemic A(H1N1) virus in late April, this virus, which had been responsible for the unseasonal outbreaks of influenza in Mexico during February and March, steadily increased in prevalence in North America, peaking during June in the USA, and spreading to other parts of the world.

In Europe, the virus was initially detected in late April (week 18) and peak detections, sentinel and non-sentinel, were recorded in the second half of July (weeks 29-31) (Figures 1 and 2). As of 14th September, ECDC reported 50,778 confirmed cases with 136 deaths (Figure 2d). The UK experienced the highest incidence of pandemic H1N1 infection, with 13,322 confirmed cases recorded, as of 9th September, with 76 associated deaths (Figure 3c). All four UK national sentinel influenza schemes showed unseasonal rises in ILI consultation rates, peaking in late July, with levels in England and Wales being higher than those for the 2008/09 season (Figure 3a), although not exceeding the epidemic activity threshold (Figure 3b). The proportion of specimens testing positive for pandemic H1N1 reached 30%, compared to 60% positive for influenza at the peak of the 2008/09 season (Figure 3d). Few seasonal influenza viruses were detected during circulation of the pandemic H1N1 virus between June and September (Figure 3d).

From August, marked increases in the prevalence of the pandemic H1N1 virus occurred in many African and Asian countries, including for example China, Japan and Kenya.

During the winter season in the southern hemisphere, influenza was due mainly to the pandemic H1N1 virus where it was prevalent in South American countries, Australia and New Zealand during June and July, and largely replaced previously circulating seasonal H1N1 and H3N2 viruses. Pandemic H1N1 virus emerged somewhat later in South Africa during August, following widespread outbreaks due to H3N2 in June.

Antigenic and genetic characteristics of pandemic A(H1N1) viruses and seasonal A(H1N1), A(H3N2) and B viruses isolated during February to August 2009

Of approximately 300 viruses isolated during February to April, the majority (54%) were A(H3N2) and 10% were A(H1N1), with influenza B accounting for 35% of the viruses characterized (Table 1). Following detection of the pandemic H1N1 virus at the end of April, it has been predominant among clinical samples and 428 viruses isolated from May to August. During the latter period, H3N2 viruses accounted for 23%, while the proportions of seasonal H1N1 (5%) and B (10%) viruses also decreased (Table 2). B/Victoria lineage viruses accounted for the majority (88%) of B viruses isolated from Feb to July.

Pandemic A(H1N1) viruses

With few exceptions, **HI tests** showed that the pandemic H1N1 viruses continued to be antigenically similar to the prototype vaccine virus A/California/7/2009 (Table 3). Many of the **HA and NA sequences** of recent isolates possess the common changes S203T in HA and V106I and N248D in NA, relative to the vaccine virus (Figures 4 and 5).

While all viruses tested were resistant to amantadine/rimantadine, sporadic detections of **oseltamivir resistance**, due to the H275Y mutation (Figure 5), have been reported; all such viruses retain sensitivity to zanamivir.

Seasonal A(H1N1) viruses

Seasonal H1N1 viruses continued to circulate at low levels. In **HI tests**, most were shown to be antigenically closely related to recent A/Brisbane/59/2007-like reference viruses, such as A/Seychelles/2239/2008 (clade 2B) or A/Hong Kong/1856/2008 (clade 2C), although HI titres were generally reduced, relative to homologous, with the antiserum to egg-grown A/Brisbane/59/2007 (Table 4). HAs of clades 2B and 2C have, in general, been indistinguishable in HI tests.

The **HA and NA sequences** of the majority of recent isolates fell within clade 2B and variation in amino acid sequence was similar to that observed earlier in 2009 (Figures 6 and 7). Different subgroups of HA were distinguished by the amino acid changes S141N and G185A, N183S and G185S, or H192R/N, but were not distinguishable in HI tests (Table 4).

The majority of viruses (clade 2B) tested were resistant to oseltamivir, retaining the H275Y mutation in NA (Figure 7), and were sensitive to zanamivir and amantadine/rimantadine.

A(H3N2) viruses

Most A(H3N2) viruses gave higher HA titres with guinea pig RBCs than with turkey RBCs and many failed to agglutinate the turkey cells. Most recent HI tests therefore used guinea pig RBCs. **HI tests** with the viruses which did not agglutinate turkey RBCs showed that many were antigenically closely related to the vaccine viruses A/Brisbane/10/2007 and A/Uruguay/716/2007, whereas others were antigenically distinct and closely related to recent reference viruses, such as A/Hong/Kong/1985/2009 and A/Perth/16/2009 (Table 5). As previously observed, most of the viruses that agglutinated both turkey and guinea pig RBCs gave reduced HI titres, relative to homologous, as shown for the MDCK cell-grown reference viruses A/Trieste/25c/2007, A/Finland/9/2008 and A/Johannesburg/15/2008 in Table 5.

Oseltamivir was observed to reduce (or abolish) haemagglutination by many MDCK cell-grown viruses, especially of turkey RBCs, and it was apparent that attachment by the NA contributed to haemagglutination. Studies of reverse genetics derived viruses have recently shown that this is due to the presence of a D151G mutation causing 'irreversible' binding to the RBCs. Addition of oseltamivir (20 nM) to HI tests with guinea pig RBCs abolished NA-dependent agglutination and allowed analysis of haemagglutination inhibition due specifically to anti-HA antibody and a clearer antigenic distinction between Brisbane/10/2007-like viruses and the recently emergent antigenic variant (Tables 6 and 7).

Virus neutralization tests also showed that A/Hong/Kong/1985/2009 was antigenically distinct from A/Brisbane/10/2007-like viruses (Table 8).

Overall, many of the viruses were shown to be antigenically and genetically closely related to A/Brisbane/10/2007 and A/Uruguay/716/2007 vaccine strains, while an increasing proportion of recent isolates were antigenically and genetically closely related to the emergent antigenic variant, represented by A/Hawaii/7/2009, A/Hong Kong/1985/2009 and A/Perth/16/2009 reference viruses.

The **HA sequences** of the latter viruses formed a separate clade characterized by the amino acid substitutions E62K, N144K, K158N and N189K (plus V213A, or I260M and R261Q) (Figure 8). The locations of these substitutions on the surface of the HA molecule is illustrated in Figure 9. There were no corresponding distinctive amino acid changes in the **NA sequences** of these viruses that distinguished them from the A/Brisbane/10/2007-like viruses (Figure 10). Viruses distinguished by the substitutions K158N, N189K and T212A in HA were also antigenically closely related to A/Hong Kong/1985/09 (Table 6); their NAs were characterized by the changes I26T and Y40C in the stalk of the molecule. Several of the recent A/Brisbane/10/2007-like viruses had acquired other changes in HA, K158R plus V112I, or P103S, A138S and T328N, and in NA, S367N and K369T.

All the viruses tested possessed asparagine 31 in M2, which confers resistance to amantadine and rimantadine. No resistance to NA inhibitors was detected.

B viruses

Influenza B viruses were predominantly of the B/Victoria-lineage.

The majority of **B/Victoria-lineage** viruses were antigenically closely related to the vaccine strain B/Brisbane/60/2008, although viruses from Hong Kong and Singapore, in particular, were closely related to the reference virus B/Hong Kong/45/05 (B/Malaysia/2506/04-like) (Table 9).

HA sequences of most of the former fell within the Brisbane/60 clade and were similar to those of earlier B/Brisbane/60/2008-like viruses, whereas those of the latter viruses were closer to the sequences of earlier B/Malaysia/2506/2004-like viruses and Chinese isolates, such as B/Jiangsu-Gulou/125/2009 or B/Hubei-Songzi/51/2008, which lack the N165K substitution (Figure 11). As previously, the **NA sequences** fell within corresponding clades and subclades of the B/Victoria-lineage (Figure 12).

The few **B/Yamagata-lineage** viruses were shown to be antigenically closely related to the previous B/Brisbane/3/2007 vaccine strain and/or more recent reference viruses, such as B/Bangladesh/3333/2007 or B/Valladolid/18/2008 (Table 10).

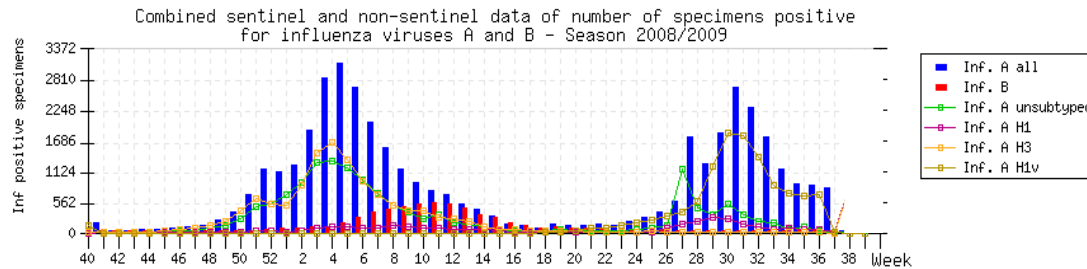
Most of their **HA and NA sequences** belonged to the Bangladesh/3333/07 clade (Figures 12 and 13).

A(H5N1) viruses

No human isolates were received. Sequences of the 8 genes of two chicken viruses from Egypt, isolated during February, showed that they were similar to those of other contemporary avian and human clade 2.2 isolates, as shown for HA and NA in Figures 14 and 15. The HAs of these recent Egyptian isolates were characterized by the amino acid substitutions D43N, S120N, I151T, P235S and S537F and a deletion of residue 129. The NAs were characterized by the substitutions L224M and S450G (plus additional substitutions R111K or Q313K and V321I, or P340S and V389G).

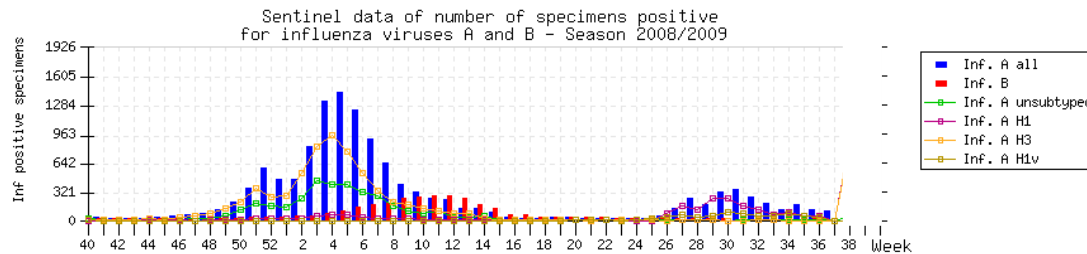
No drug resistance mutations were detected in the NA or M2 proteins.

Figure 1: Influenza positive specimens 2008/9 season - Europe.



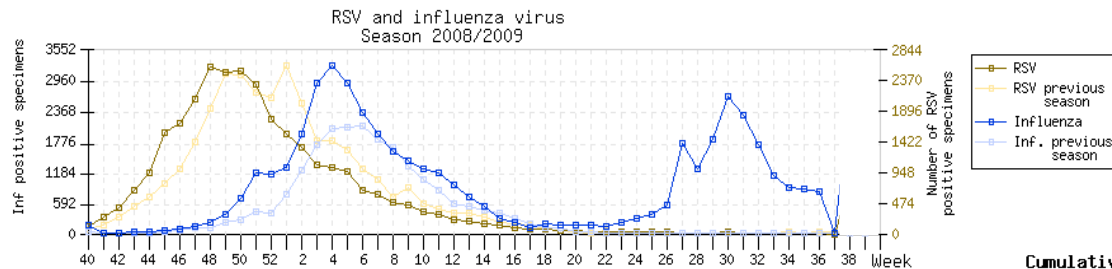
Created at 00:32 on Sep 15 2009

Source: The WHO European Influenza Network



Created at 00:32 on Sep 15 2009

Source: The WHO European Influenza Network



Created at 00:32 on Sep 15 2009

Source: The WHO European Influenza Network

Snapshot taken at week 36/2009.

Season effectively over by week 16/2009.

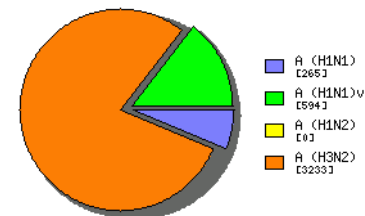
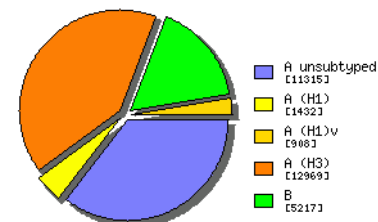
Predominantly an A(H3N2) season.

Pandemic H1N1 picked up initially in non-sentinel sampling.

Cumulative influenza virus isolate typings and subtypings* Season 2008/2009

Total number of isolates - IN = 31841
(includes N-subtyped and N-unsubtyped isolates)

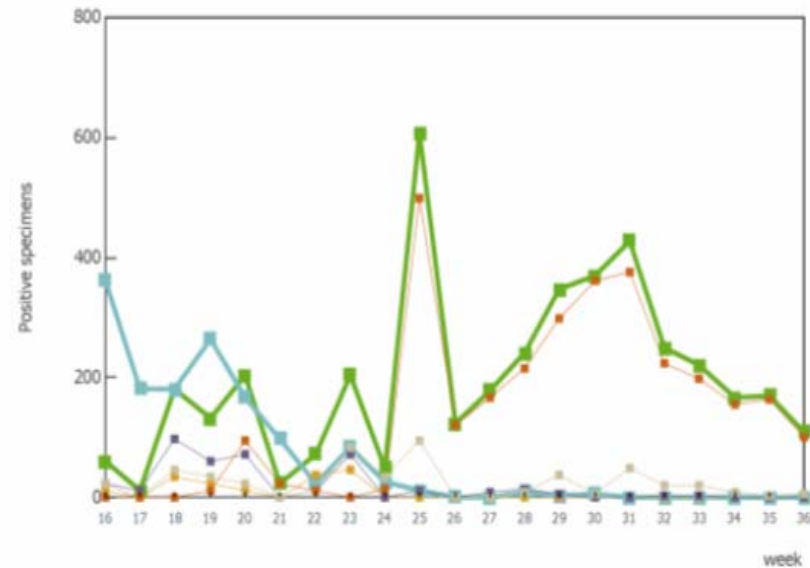
Number of isolates N-subtyped - IN = 40921



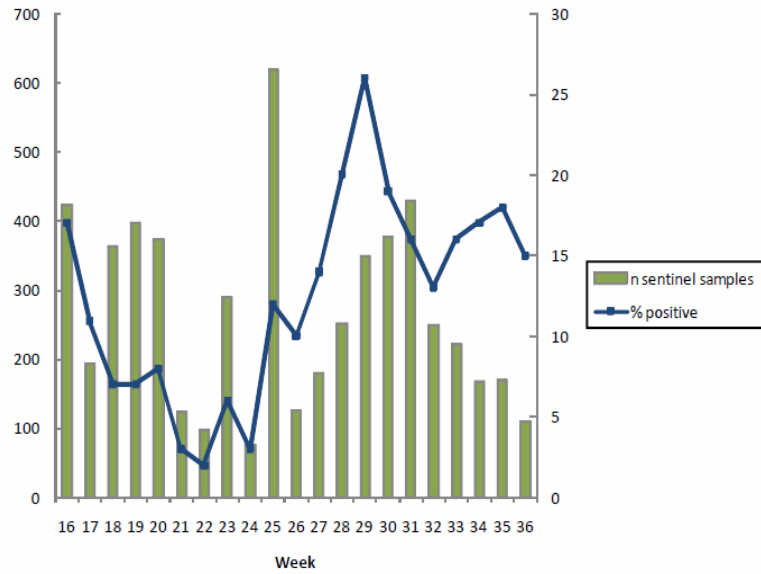
Data taken from EuroFlu.org

Figure 2: Pandemic (H1N1) 2009 virus detections – Europe (ECDC data).

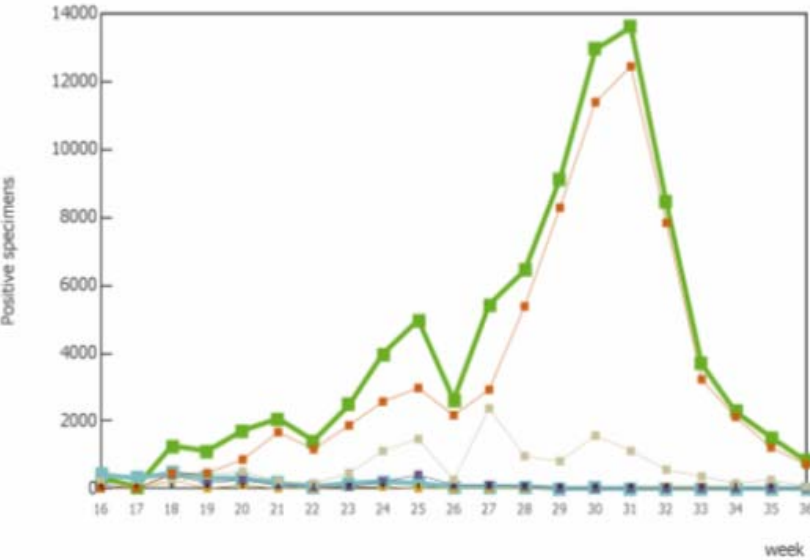
a) Number of sentinel specimens positive for influenza by type, subtype and week of report.



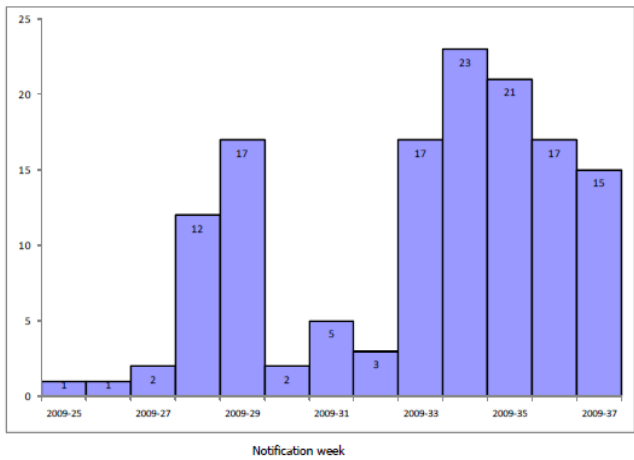
b) Proportion of sentinel samples positive for influenza by week of report.



c) Number of non-sentinel specimens positive for influenza by type, subtype and week of report.



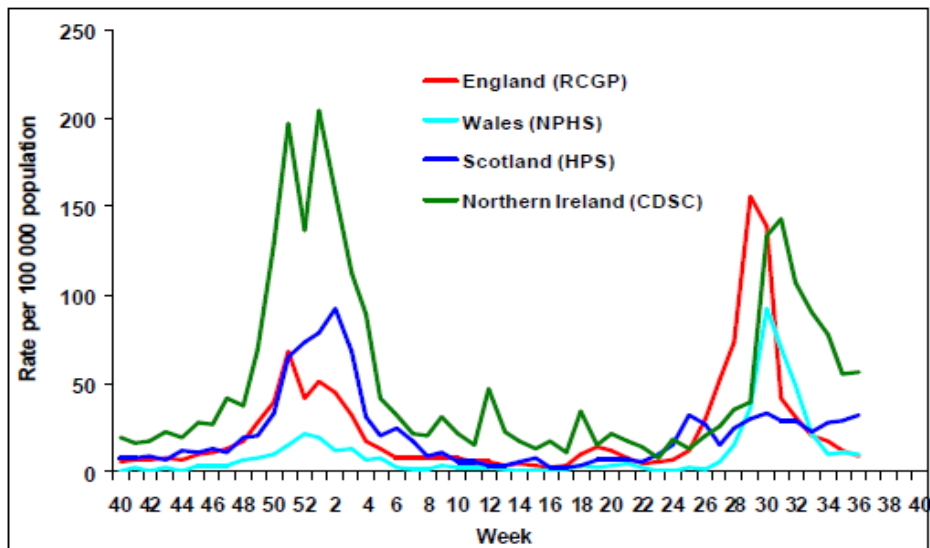
d) Number of confirmed deaths among pandemic (H1N1) 2009 influenza cases by week of notification: EU and EFTA countries.



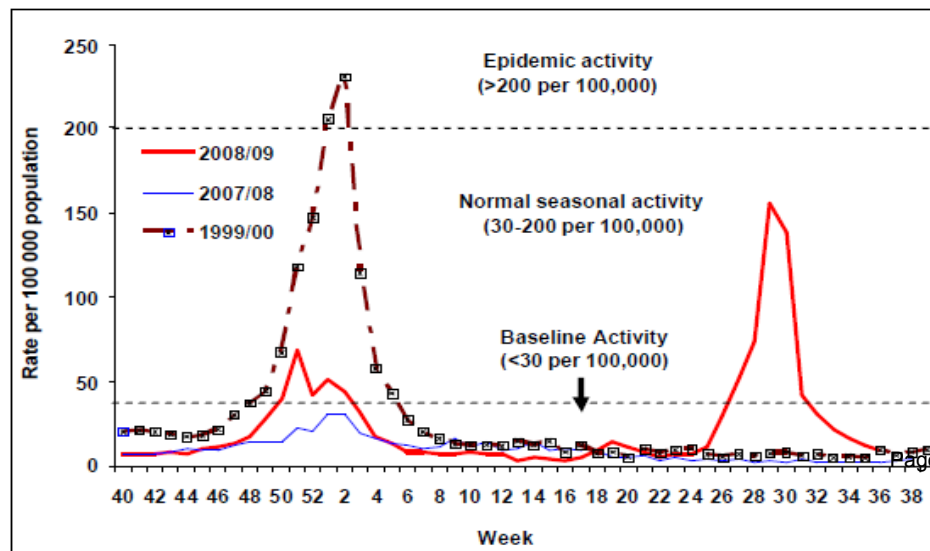
As of 14.09.09: 50778 confirmed cases, 136 deaths.

Figure 3: HPA (UK) National Influenza Report, 10.09.09 (week 37).

a) GP weekly consultation rates for influenza/ILI in the UK national sentinel influenza schemes, 2008/09.



b) RCGP weekly consultation rate for ILI 2008/09 and recent years.

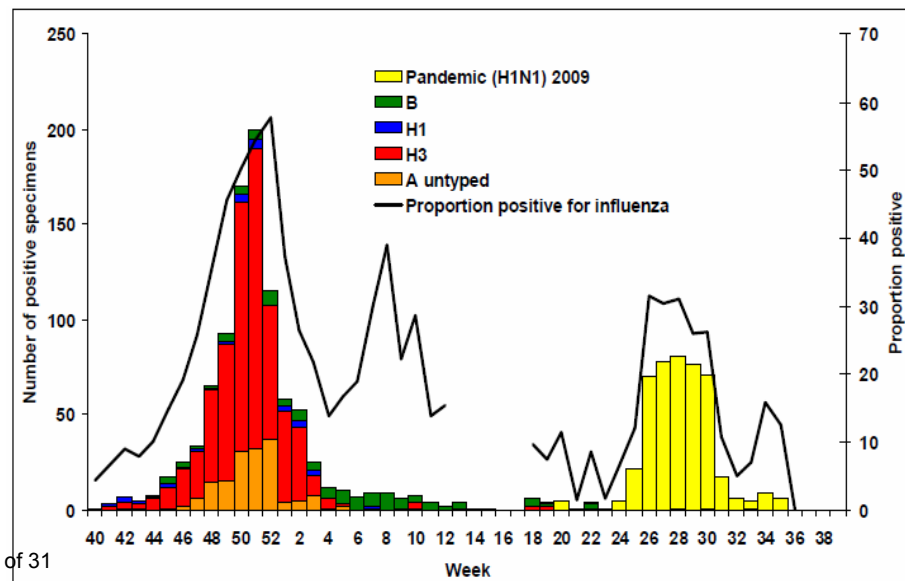


c) Cumulative number of laboratory confirmed cases, as of 09.09.09.

Region / Country	Cumulative number of laboratory confirmed cases
England	11,155
Northern Ireland	189
Scotland	1858
Wales	120
Total UK *	13,322

* 76 deaths.

d) Number of samples testing positive for influenza in the two GP-based English sentinel virological schemes.



NB. Proportion positive omitted when fewer than 10 specimens reported; data for the most recent weeks are subject to change due to reporting lag.

Table 1. Geographic origins of human influenza isolates characterised, February 2009 - August 2009

Country	H1N1		H3N2		B		
	A/Bris	A/Cal	A/Bris	A/HK	Yamagata	Victoria	
	59/07	7/09	10/07	1985/09	lineage FI/4/06	Bris/60	HK/45/05
February 2009							
Albania			2				
Belgium	1		3			5	
Cameroon	1				1		
Croatia						1	
Czech Republic	1					2	
Egypt					1		
Estonia			3			1	
France	2		10		1	4	3
Georgia			5			1	
Ghana					1		
Greece			2			1	
Iran			1		2		1
Italy	4		5			1	
Latvia	1		5		1	4	
Luxembourg			5				
Madagascar			2				
Netherlands	1					1	
Norway	1		1				
Russia	1		3				
Serbia			3				7
Spain						2	
Sweden	1		5				
Switzerland			4				
Ukraine			15			1	
March 2009							
Croatia			1			4	
Egypt					2		
Estonia			3				
Finland	1					7	
Georgia	1		4				
Hong Kong	3		3		1		
Iran	2						
Luxembourg			1				
Madagascar			3			4	
Moldova			3				
Norway			3				
Russia	2		11			5	
Serbia							5
Singapore	1						
Sweden			1				
Switzerland			1			2	
Ukraine			27			3	
April 2009							
Croatia			4				
Egypt	1				3		
Finland	1		2			2	
France			1				
Georgia						4	
Ghana			2		1		
Hong Kong	1		1				2
Madagascar			5			1	
Netherlands			1			1	
Norway	2					3	
Russia	1		4			1	
Singapore			1				3
Spain		2					
South Africa			1				
Ukraine			1			6	
Subtotal = 292	30 10%	2 1%	158 158 (54%)		14	67 102 (35%)	21

Country	H1N1		H3N2		B		
	A/Bris	A/Cal	A/Bris	A/HK	Yamagata	Victoria	
	59/07	7/09	10/07	1985/09	lineage FI/4/06	Bris/60	HK/45/05
May 2009							
Belgium		7					
Finland		2					
France	1	3	1				
Germany		3					
Georgia							1
Ghana	4						
Greece		2					
Hong Kong				4			
Iran					1		
Ireland		7					
Israel		1					
Italy		4					
Luxembourg			3				
Madagascar			2				5
Norway		7	1			1	11
Romania		1					
Singapore	2			5			1
Spain		21					
Sweden					1		
South Africa			15	1			
Ukraine		1					
United Arab Emirates		1					
United Kingdom		6					
June 2009							
Algeria		4					
Austria		4					
Belgium		3					
Bulgaria		2					
Cameroon			13				
Cyprus		6					
Denmark		3					
Estonia		2					
France	1			1			
Ghana	8		1				2
Hong Kong	1						
Ireland		2					
Italy		6					
Latvia		1					
Luxembourg		3					
Macedonia		2					
Madagascar			2				9
Malta		4					
Mauritius		2	2				
Morocco		4					
Netherlands			1				
Norway		8		2			1
Portugal		8	1				
Singapore				2			
Senegal				2			
Serbia and Montenegro		1					
Singapore		5					
Slovenia		1					
South Africa		4	10	1			
Sweden							1
United Arab Emirates		3					
Qatar		5					
July 2009							
Austria		13					
Bulgaria		9					
Cameroon			10				
Ghana			3	1			4
Greece		1					
Hong Kong	3	3		4		1	1
Italy		2					
Latvia		5					
Macedonia		1					
Malta		64					
Mauritius				4			2
Moldova							
Portugal		5					
Slovenia		3					
United Kingdom			3	4			
August 2009							
Cameroon		2	2				
Luxembourg		4					
Mauritius		4					
Subtotal = 428	20 5%	265 62%	70 101 (23%)	31	4	36 42 (10%)	2

Table 2. Summary of human influenza isolates characterised, February 2009 - August 2009

Month of Collection	H1N1		H3N2		B		
	A/Bris 59/07	A/Cal 7/09	A/Bris 10/07	A/HK 1985/09	Yamagata lineage Fl/4/06	Victoria lineage Bris/60	HK/45/05
February 2009	14	0	74	0	7	24	11
March 2009	10	0	61	0	3	25	5
April 2009	6	2	23	0	4	18	5
May 2009	7	66	22	10	3	17	1
June 2009	10	83	30	8	0	13	0
July 2009	3	106	16	13	1	6	1
August 2009	0	10	2	0	0	0	0
Total = 720	50 7%	267 37%	228 259 (36%)	31	18	103 144 (35%)	23

Table 3. Antigenic analyses of pandemic influenza A(H1N1) viruses

Haemagglutination inhibition titre ¹												
Viruses	Collection date	Passage History	Post infection ferret sera									
			A/NJ ²	A/sw/Iowa	A/NJ	A/Cal	A/Cal	A/Eng	A/Auck	A/Narita	A/Ohio	A/Bris
			8/76 R15/79	15/30 F31/99	8/76 F34/82	4/09 C4/F14/09	7/09 C4/16/09	NIBSC F17/09	3/09 C4/17/09	1/09 JAP 6/09	2/07 CDC 97/08	59/07 F5/08
REFERENCE VIRUSES												
A/Iowa/15/30		Ex	2560	1280	<	<	<	<	<	<	<	<
A/New Jersey/8/76		Ex	>5120	160	640	320	640	320	640	1280	1280	<
A/Ohio/2/2007		C3 / MDCK1	2560	<	320	1280	1280	2560	5120	5120	1280	<
A/California/4/2009		C1,M1,E1	5120	40	320	2560	2560	2560	5120	5120	5120	<
A/California/7/2009		E1+4	2560	<	160	1280	1280	1280	2560	5120	5120	<
A/England/195/2009		MDCK3	2560	<	80	640	1280	1280	1280	5120	5120	<
A/Auckland/3/2009		Ex+2	5120	40	320	2560	2560	2560	5120	5120	5120	<
A/Narita/1/2009		E2+2	5120	40	320	2560	2560	2560	5120	5120	5120	<
A/Brisbane/59/2007		E7	<	<	<	<	<	<	<	<	<	320
TEST VIRUSES												
A/Osaka/180/2009		Cx /MDCK1	2560	<	80	1280	1280	1280	1280	5120	5120	<
A/Brandenburg/19/2009	02/05/2009	Siat3,MDCK1	2560	<	<	320	320	160	160	320	160	<
A/CastillayLeon/321/2009	23/05/2009	MDCK1/MDCK4	1280	<	80	1280	1280	640	320	1280	ND	<
A/Lyon/1041/2009	28/05/2009	MDCK2+1	2560	<	160	1280	2560	640	2560	2560	2560	<
A/Ukraine/ 259/2009	29/05/2009	MDCK4	2560	<	160	1280	2560	1280	640	2560	ND	<
A/Belgium/ 182/2009	31/05/2009	MDCK2	2560	<	80	1280	1280	1280	320	1280	ND	<
A/Abu Dhabi/718123orig/2009	Jun-09	MDCK4	5120	<	160	2560	2560	2560	2560	>5120	5120	<
A/Cyprus/S14/2009	Jun-09	MDCK3	2560	<	160	1280	1280	640	2560	2560	2560	<
A/Ireland/35770/2009	Jun-09	MDCK3	1280	<	80	1280	2560	1280	640	1280	ND	<
A/Macedonia/64/2009	Jun-09	MDCK3	1280	<	80	1280	1280	1280	1280	5120	5120	<
A/Slovenia/2312/2009	Jun-09	MDCK2	2560	<	160	2560	2560	2560	2560	5120	5120	<
A/Norway/1731/2009	08/06/2009	MDCK1+2	5120	<	160	2560	2560	1280	2560	>5120	2560	<
A/Denmark/528/09 Res	09/06/2009	sw kidney2, MDCK3	2560	<	80	2560	1280	1280	640	2560	ND	<
A/Qatar/89/2009	17/06/2009	MDCK3	1280	<	80	1280	1280	1280	2560	2560	2560	<
A/Singapore/99/2009	17/06/2009	MDCK 1+1	2560	<	160	1280	1280	640	2560	2560	2560	<
A/Dakar/011/09	18/06/2009	MDCK2	2560	<	160	1280	1280	1280	2560	2560	2560	<
A/Montenegro/2816/2009	21/06/2009	MDCK2	2560	<	160	1280	1280	1280	2560	>5120	2560	<
A/Luxembourg/871/2009	23/06/2009	MDCK1+3	2560	<	160	2560	2560	2560	2560	>5120	5120	<
A/Marrakech/47/2009	23/06/2009	MDCK2	1280	<	160	1280	1280	1280	1280	5120	5120	<
A/Estonia/28FA/2009	24/06/2009	MDCK4	1280	<	80	640	1280	640	2560	5120	5120	<
A/South Africa/4/2009	24/06/2009	MDCK3	1280	<	80	1280	640	1280	1280	5120	5120	<
A/Norway/2003/2009	26/06/2009	MDCK1/MDCK1	2560	<	80	1280	1280	1280	2560	5120	5120	<
A/Algeria/G625/2009	28/06/2009	MDCK3	1280	<	160	1280	2560	1280	2560	5120	5120	<
A/Greece/366/2009	Jul-09	MDCK3	2560	<	160	2560	1280	1280	2560	5120	5120	<
A/Malta/MV566541/2009	Jul-09	MDCK2	2560	<	160	2560	1280	1280	2560	5120	5120	<
A/Lisboa/39/2009	01/07/2009	MDCK2/MDCK1	2560	<	80	1280	1280	640	1280	5120	5120	<
A/Austria/501483/2009	06/07/2009	C1 / MDCK2	640	<	<	320	160	160	160	ND	ND	<
A/Italy/case187/09	07/07/2009	MDCK2	2560	<	160	1280	1280	1280	1280	2560	2560	<
A/Latvia/7-655/2009	11/07/2009	MDCKx/MDCK1	2560	<	160	1280	2560	1280	2560	5120	5120	<
A/Austria/502995/2009	13/07/2009	MDCK1+1	2560	<	160	1280	1280	1280	2560	2560	ND	<
A/Hong Kong/5548/2009	18/07/2009	MDCK2+1	5120	<	160	2560	2560	2560	2560	5120	5120	<
A/Bulgaria/340/2009	24/07/2009	MDCK2	1280	<	80	1280	1280	1280	1280	1280	ND	<
A/Modra/1156/2009	27/07/2009	MDCK2	1280	<	80	1280	1280	1280	1280	1280	ND	<
A/Moldova39/2009	31/07/2009	MDCK2	2560	<	160	1280	2560	1280	1280	2560	2560	<
A/Iran/2521/2009	Aug-09	E2 / E1	2560	<	160	2560	2560	1280	2560	ND	ND	<
A/Bratislava/1258/2009	09/08/2009	MDCK2	2560	<	160	1280	1280	1280	2560	2560	ND	<
A/Mauritius4AA/2009	06/08/2009	MDCK2	2560	<	160	1280	1280	1280	2560	2560	ND	<

1. < = 40; 2. hyperimmune rabbit serum

Figure 4. Phylogenetic comparison of pandemic influenza A H1N1 HA genes

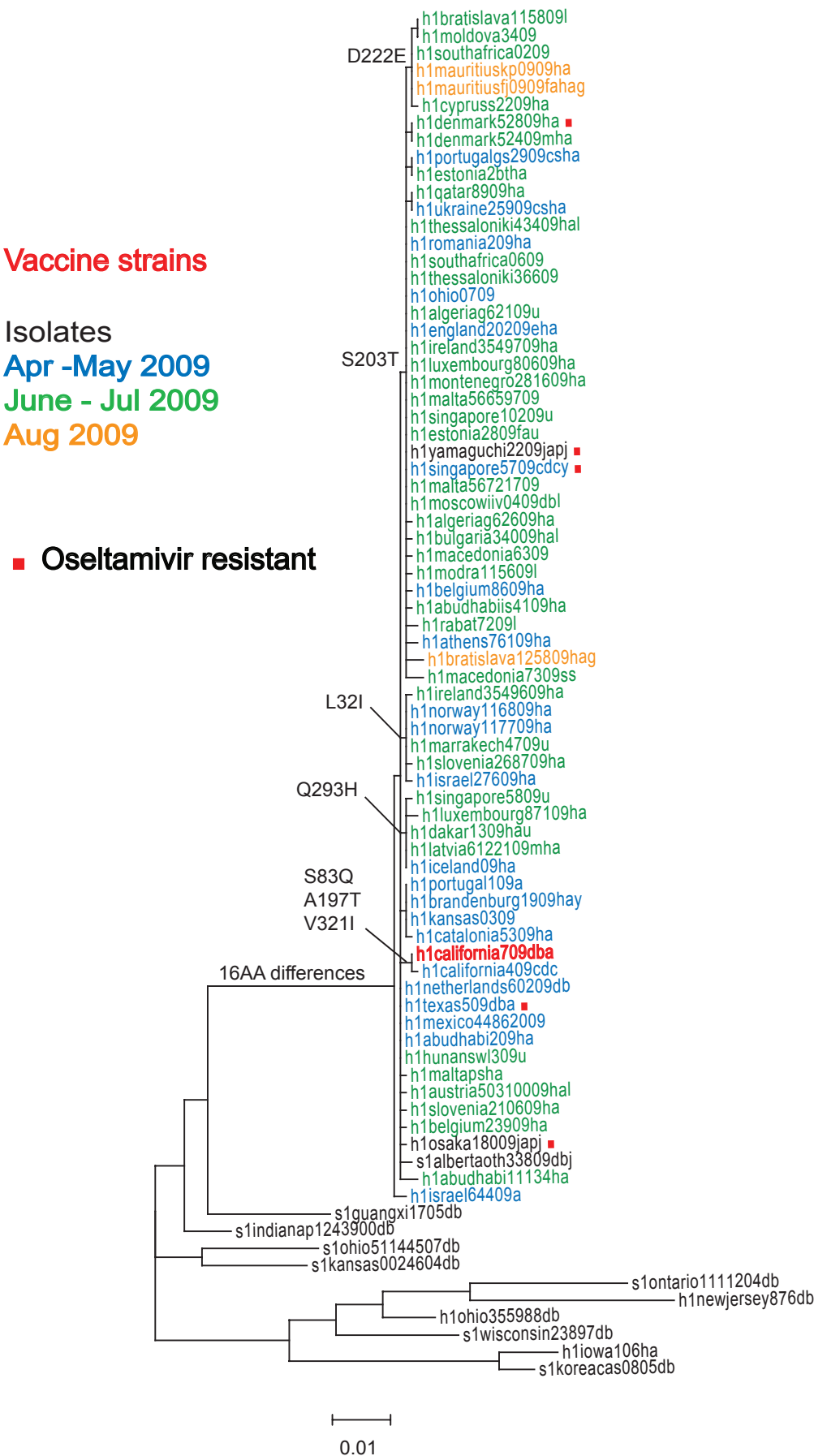


Figure 5. Phylogenetic comparison of pandemic influenza A H1N1 NA genes

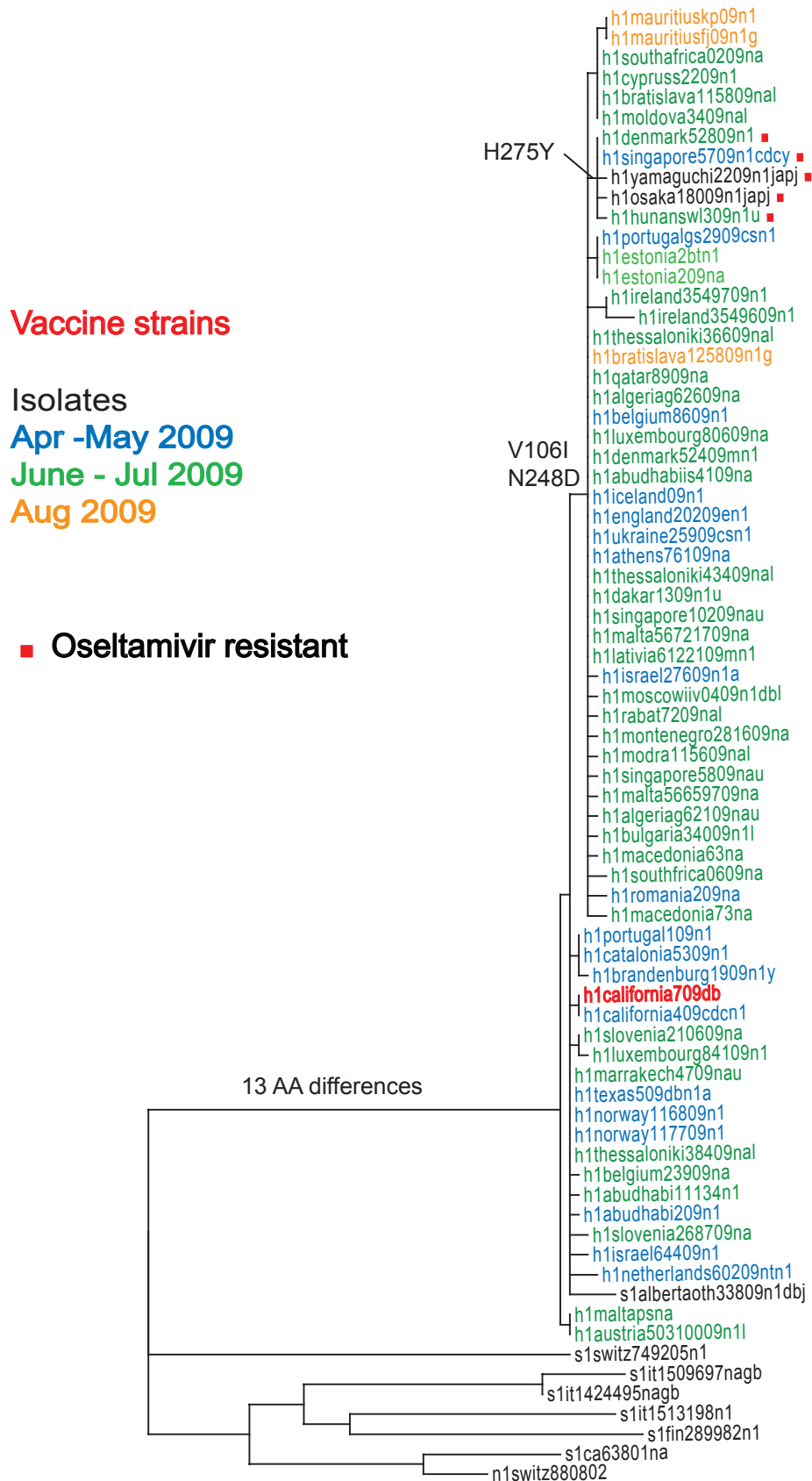


Table 4. Antigenic analyses of seasonal influenza A H1N1 viruses

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹										
			Post infection ferret sera										
			A/NC	A/HK	A/SI	A/Neth	A/Bris	A/SP	A/Sey	A/HK	A/HK	A/Mos	A/HK
			20/99	2652/06	3/06	345/07	59/07	12/08	2239/08	1856/08	1870/08	2/09	1988/09
Phylogenetic Clade			F19/08	F14/06	F8/07	F23/07	F6/08	F17/08	F27/08	F28/08	F4/09	F19/09	F20/09
			1	2C	2A	2B	2B	2B	2B	2C	2C	2B	2B
REFERENCE VIRUSES													
A/New Caledonia/20/99	06/09/1999	Ex	640	40	40	80	40	40	40	80	40	160	80
A/Hong Kong/2652/2006	17/07/2006	E2 E2	40	160	160	80	320	160	80	320	640	80	320
A/Solomon Islands/3/2006	21/08/2006	E3 E2	80	160	320	320	80	80	320	320	160	320	160
A/Netherlands/345/2007	12/10/2007	MDCK2 Siat4	80	160	160	640	160	160	320	640	160	80	160
A/Brisbane/59/2007	01/07/2007	E2 E3	40	80	160	80	320	320	80	160	320	80	160
A/St.Petersburg/12/2008	13/02/2008	E2 E2	<	80	160	80	320	320	160	160	320	160	160
A/Seychelles/2239/2008	03/06/2009	MDCK1 Siat2	<	80	80	80	40	80	160	160	160	80	80
A/Hong Kong/1856/2008	24/07/2008	MDCK2 Siat2	40	80	40	160	40	40	320	640	320	80	160
A/Hong Kong/1870/2008	26/07/2008	MDCK4 Siat2	<	160	160	160	160	160	80	320	640	80	80
A/Moscow/2/2009 ^a	23/03/2009	MDCK1 Siat2	80	80	80	1280	320	1280	160	640	640	640	160
A/Hong Kong/1988/2009 ^a	29/03/2009	MDCK2 Siat2	<	40	40	80	80	160	160	160	160	160	640
TEST VIRUSES													
A/Norway/736/2009	12/03/2009	MDCK1 Siat2	40	160	160	640	40	80	320	640	160	ND	ND
A/Singapore/10/2009 ^b	16/03/2009	MDCK 1 Siat1	160	320	160	320	160	160	320	640	320	640	640
A/Moscow/9/2009	23/03/2009	C1 Siat2	<	40	80	160	40	80	160	320	80	ND	ND
A/Hong Kong/1987/2009 ^c	28/03/2009	MDCKx2 Siat1	<	40	40	320	80	160	320	320	160	ND	ND
A/Finland/514/2009 ^a	30/03/2009	M1 Siat1	<	<	80	80	80	80	160	320	160	ND	ND
A/Hong Kong/1968/2009	01/04/2009	MDCKx2 Siat1	<	80	<	160	80	80	160	160	80	ND	ND
A/Finland/537/2009	02/04/2009	M1 Siat1	<	80	80	160	<	80	160	640	160	ND	ND
A/St. Petersburg/66/2009	06/04/2009	E1 Siat1	<	80	40	<	40	160	40	40	160	ND	ND
A/Menya/1386/2009 ^c	12/04/2009	x Siat1	<	<	40	80	40	40	160	320	80	80	80
A/Norway/998/2009	29/04/2009	MDCK1 Siat2	<	80	80	40	80	160	80	80	160	80	320
A/Singapore/37/2009	02/05/2009	MDCK 1 Siat1	<	80	80	40	80	160	40	40	160	80	320
A/Ghana/FS-09-718/2009 ^a	08/05/2009	x Siat2	<	40	<	160	40	40	160	640	80	320	320
A/Singapore/35/2009	08/05/2009	MDCK1 Siat1	40	160	160	640	160	320	320	640	160	1280	1280
A/Aurillac/974/2009	09/05/2009	MDCK2 Siat1	<	40	40	160	40	40	160	160	80	80	160
A/Ghana/FS-09-749/2009	15/05/2009	x Siat2	<	<	80	160	40	40	160	640	160	160	160
A/Ghana/FS-09-791/2009 ^a	21/05/2009	x Siat3	40	160	80	320	80	80	320	1280	160	640	1280
A/Singapore/52/2009	21/05/2009	MDCK1 Siat2	<	<	<	160	<	40	80	160	40	640	320
A/Ghana/FS-09-881/2009	08/06/2009	x Siat2	40	80	40	320	40	80	160	640	160	640	1280
A/Iceland/8599/2009	18/06/2009	MDCK2 Siat1	<	<	40	80	40	80	80	160	160	40	80
A/Ghana/FS-09-978/2009	30/06/2009	x Siat1	40	80	80	320	80	160	320	640	160	640	1280
A/Marseille/1148/2009 ^b	18/06/2009	MDCK2 Siat1	<	40	80	160	160	80	160	320	160	80	320
Algeria/G644/2009	Jul-09	Cx Siat2	40	80	80	160	80	80	160	640	160	640	640
A/Hong Kong/4278/2009	04/07/2009	MDCK2 Siat1	<	40	40	160	40	80	80	320	40	320	320

1. < = <40; ND = not done

a. S141N, G185A; b. N183S, G185S; c. H192N

Vaccine strain

Figure 6. Phylogenetic comparison of seasonal influenza H1N1 HA genes

Vaccine strains

Reference strains

Isolates

Jan - Feb 2009

Mar - Apr 2009

May - Jun 2009

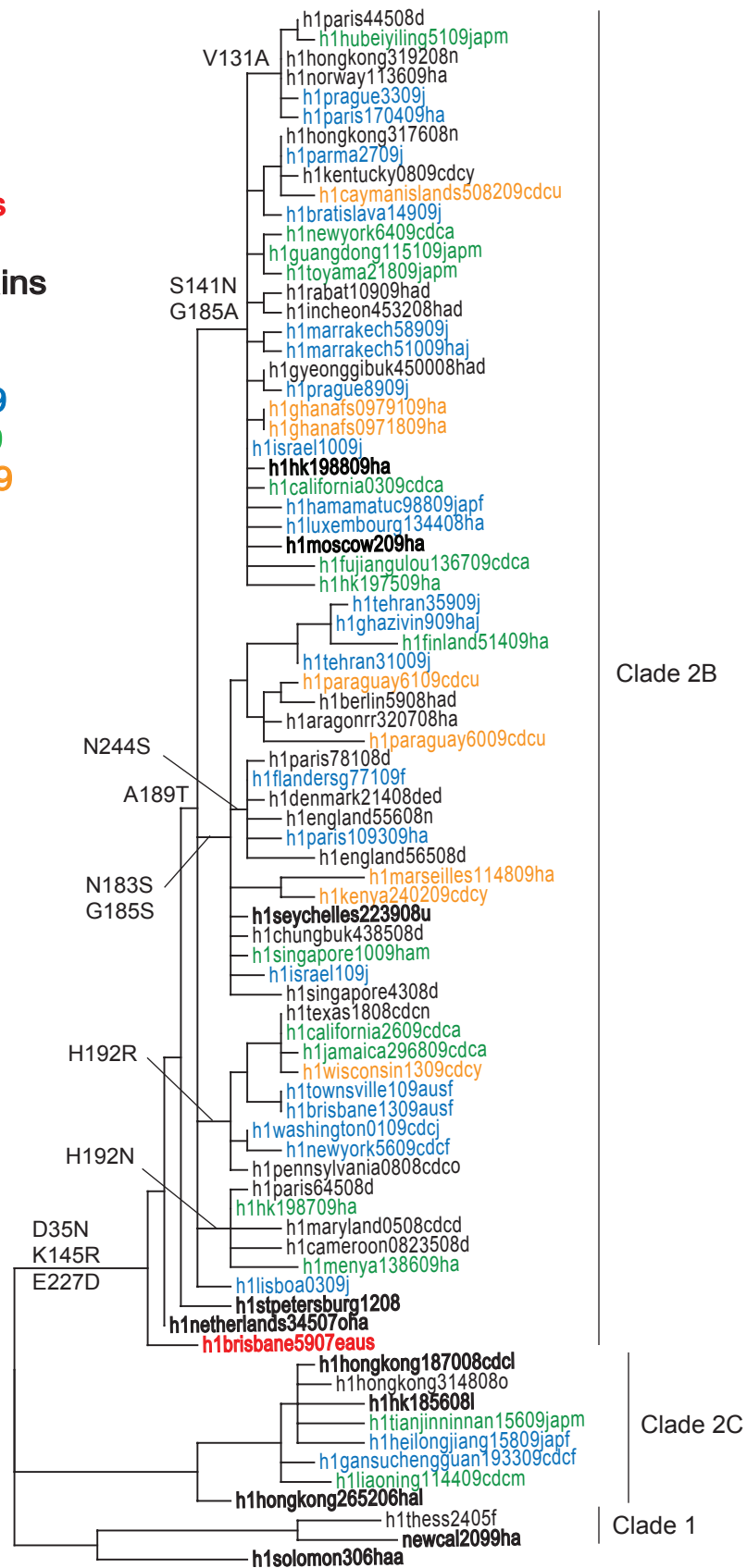


Figure 7. Phylogenetic comparison of seasonal influenza H1N1 NA genes

Vaccine strains

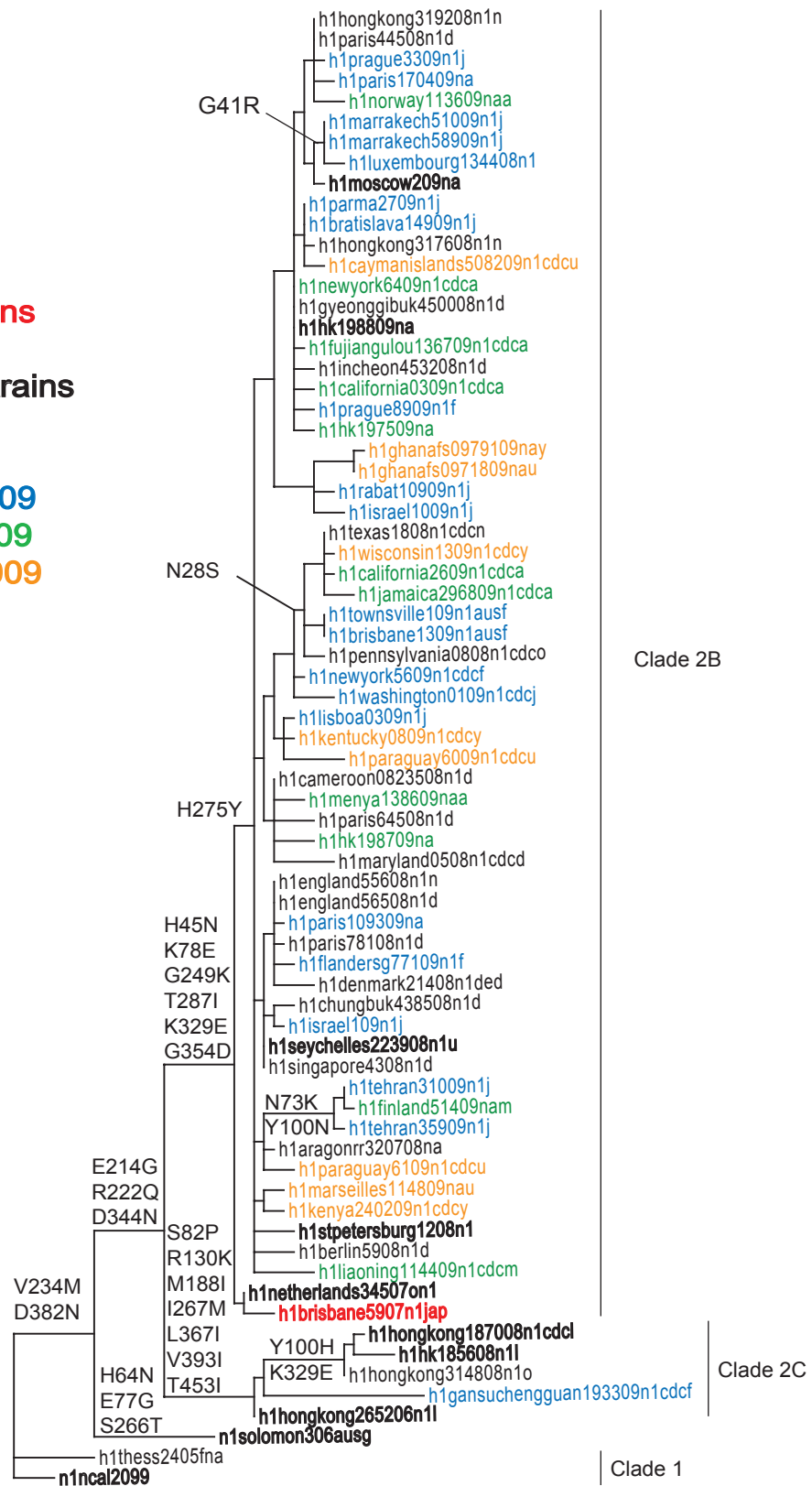
Reference strains

Isolates

Jan - Feb 2009

Mar - Apr 2009

May - Jun 2009



0.01

Table 5. Antigenic analyses of influenza A H3N2 viruses - Guinea Pig RBC

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹										
			Post infection ferret sera										
			A/Wis	A/Trieste	A/Bris	A/Uru	A/Fin/	A/Jhb	A/Bris	A/HK	A/HK	A/Perth	A/Perth
			67/05	25c/07	10/07	716/07	9/08	15/08	24/08	1952/09	1985/09	15/09	16/09
			F1/06	F5/07	F29/08	F26/08	F12/08	F22/08	F3/09	F22/09	F21/09 Aus	F1412 Aus	F1413
REFERENCE VIRUSES													
A/Wisconsin/67/2005	31/08/2005	SpfCk3E3 E3	1280	1280	640	2560	640	640	1280	<	40	<	<
A/Trieste/25c/2007	Jan-07	MDCKx Siat4	<	80	80	80	40	40	80	40	80	40	40
A/Brisbane/10/2007	06/02/2007	E2 E3	1280	2560	2560	5120	2560	1280	2560	40	80	40	40
A/Uruguay/716/2007	21/06/2007	spfck1, E3 E1	640	640	640	1280	640	320	640	<	<	<	<
A/Finland/9/2008	07/01/2008	MDCK2 Siat2	160	320	640	320	640	320	320	80	320	160	160
A/Johannesburg/15/2008	25/06/2008	MDCKx Siat2	160	320	320	320	320	320	320	80	160	160	160
A/Brisbane/24/2008	23/06/2008	E5 E1	1280	2560	5120	5120	1280	1280	2560	<	40	<	40
A/Hong Kong/1952/2009	24/03/2009	MDCKx2 Siat2	160	160	320	320	320	160	320	640	2560	1280	1280
A/Hong Kong/1985/2009	01/04/2009	MDCKx2 Siat2	40	40	80	160	40	40	40	160	640	320	320
A/Perth/15/2009		E3 E1	<	<	<	<	<	<	<	80	640	320	320
A/Perth/16/2009		E3 E1	<	<	<	<	<	<	<	160	640	320	640
TEST VIRUSES													
A/Madagascar/4007/2009	26/05/2009	MDCK1 Siat1	160	320	320	1280	320	320	320	80	160	40	80
A/Madagascar/4010/2009	28/05/2009	MDCK1 Siat1	160	640	640	1280	320	640	320	80	320	80	160
A/Madagascar/4286/2009	02/06/2009	MDCK1 Siat1	160	160	320	640	160	160	320	80	320	80	80
A/Ukraine/86/09	02/03/2009	MDCK1 Siat1	640	640	1280	1280	640	640	640	40	80	ND	ND
A/Odessa/366/09	26/03/2009	MDCK1 Siat1	320	640	1280	1280	640	640	640	40	80	ND	ND
A/Luxembourg/612/2009	19/02/2009	Siat2	640	640	1280	1280	320	640	640	40	40	ND	ND
A/Johannesburg/131/2009	23/05/2009	MDCK1 Siat1	1280	1280	2560	5120	1280	1280	1280	320	320	320	80
A/Johannesburg/161/2009	26/05/2009	MDCK1 Siat1	640	1280	1280	1280	640	640	640	80	160	160	160
A/Johannesburg/239//2009	02/06/2009	MDCK1 Siat1	640	1280	640	1280	640	180	1280	40	80	80	80
A/Lyon/CHU/20.526/2009	14/05/2009	MDCK2 Siat1	80	320	160	320	160	320	160	<	40	<	<
A/Hong Kong/2139/2009	16/05/2009	MDCKx2 Siat1	160	160	160	640	160	160	160	640	5120	1280	1280
A/Hong Kong/2142/2009	17/05/2009	MDCKx2 Siat1	80	80	160	640	80	80	80	640	2560	640	1280
A/Singapore/46/2009	05/06/2009	MDCK 1 Siat1	40	40	80	320	40	40	80	320	5120	640	640
A/Singapore/47/2009	05/06/2009	MDCK 1 Siat1	80	40	80	160	40	40	80	640	5120	1280	1280
A/Grenoble/1144/2009	17/06/2009	MDCK2 Siat1	80	640	160	320	160	320	160	1280	5120	180	2560
A/Luxembourg/782/2009	08/05/2009	MDCK1 Siat1	<	40	80	80	40	40	40	320	640	320	640
A/Luxembourg/791/2009	11/05/2009	Siat3	40	40	80	160	<	40	40	640	2560	ND	ND
A/Singapore/42/2009	21/05/2009	MDCK 1 Siat1	80	320	160	160	40	80	80	640	2560	1280	1280
A/Johannesburg/126/2009*	21/05/2009	MDCK2 Siat1	320	640	1280	640	640	<	160	320	160	640	1280

1. < = <40

Vaccine strains

Table 6. Antigenic analyses of influenza A H3N2 viruses - Guinea Pig RBCs - 20nM Oseltamivir

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹								
			Post infection ferret sera								
			A/Wis	A/Trieste	A/Bris	A/Uru	A/Fin/	A/HK	A/HK	A/Perth	A/Perth
			67/05	25c/07	10/07	716/07	9/08	1952/09	1985/09	15/09	16/09
			F18/08	F5/07	F29/08	F26/08	F12/088	F22/09	F21/09 Aus	F1412 Aus	F1413
REFERENCE VIRUSES											
A/Wisconsin/67/2005	31/08/2005	SpfCk3E3 \3	640	1280	1280	1280	320	<	<	<	<
A/Trieste/25c/2007	Jan-07	MDCKx Siat4	640	1280	1280	2560	640	40	80	40	40
A/Brisbane/10/2007	06/02/2007	E2 \3	5120	2560	5120	5120	5120	40	80	320	80
A/Uruguay/716/2007	21/06/2007	spfck1, E3 \1	1280	1280	1280	2560	1280	<	<	<	<
A/Finland/9/2008	07/01/2008	MDCK2 \2	1280	1280	2560	5120	1280	40	80	80	80
A/Hong Kong/1952/2009	24/03/2009	MDCKx2 \2	40	80	<	160	80	640	5120	640	1280
A/Hong Kong/1985/2009	01/04/2009	MDCKx2 \2	80	80	<	160	80	320	5120	640	1280
A/Perth/15/2009		E3 \1	<	<	<	<	<	160	640	320	320
A/Perth/16/2009		E3 \1	<	<	<	<	<	80	320	160	320
TEST VIRUSES											
A/Toulouse/955/2009	29/04/2009	MDCK2 Siat1	1280	2560	2560	5120	1280	<	40	40	40
A/Brisbane/54/2009	07/05/2009	E3 E1	640	1280	640	2560	640	<	40	ND	ND
A/Cape Town/294/2009	22/05/2009	MDCK2 Siat1	320	640	1280	2560	320	<	40	<	40
A/Durban/675/2009	Jun-09	MDCK2 Siat3	160	320	640	1280	160	40	40	40	40
A/Madagascar/4286/2009	02/06/2009	MDCK1 Siat1	160	160	640	640	160	80	80	40	40
A/Johannesburg/556/2009	10/06/2009	MDCK1 Siat1	640	2560	1280	2560	1280	<	40	40	40
A/Lisboa/32/2009	25/06/2009	MDCK2 Siat1	160	320	320	640	320	80	80	80	80
A/England/233/2009	01/07/2009	Cx Siat1	320	640	1280	1280	640	<	40	ND	ND
A/Ghana/FS-1035/2009	13/07/2009	MDCK1 Siat1	40	320	320	160	160	40	80	ND	ND
A/Singapore/33/2009	07/05/2009	MDCK1 Siat1	40	40	<	160	40	640	5120	640	640
A/Hong Kong/2139/2009	16/05/2009	MDCK2 Siat3	160	160	80	160	160	640	2560	640	640
A/Hong Kong/2142/2009	17/05/2009	MDCK2 Siat1	40	80	<	160	40	640	1280	640	640
A/Singapore/41/2009	19/05/2009	MDCK1 Siat1	40	40	<	160	40	640	2560	640	640
A/Singapore/37/2009	01/06/2009	E2 E1	<	<	<	<	<	80	640	ND	ND
A/England/232/2009	01/06/2009	Cx Siat1	<	40	80	80	40	640	2560	ND	ND
A/Scotland/5/2009	01/06/2009	Cx Siat1	40	40	80	80	40	320	1280	ND	ND
A/Singapore/46/2009	05/06/2009	MDCK1 Siat1	40	40	40	160	40	320	5120	640	640
A/Norway/1739/2009	10/06/2009	MDCK1 Siat1	40	80	160	60	40	640	2560	ND	ND
A/Grenoble/1144/2009	17/06/2009	MDCK2 Siat1	40	40	<	160	40	320	2560	640	640
A/Dakar/005/09	23/06/2009	MDCK2 Siat1	160	80	320	320	80	640	5120	ND	ND
A/Hong Kong/4494/2009	10/07/2009	MDCKx2 Siat1	40	40	80	160	40	640	2560	ND	ND
A/Johannesburg/126/2009*	21/05/2009	MDCK2 Siat1	320	640	1280	640	640	320	160	640	1280
A/Victoria/208/2009*	02/06/2009	E3 E1	80	<	160	160	<	320	1280	ND	ND
A/Johannesburg/385/2009*	08/06/2009	MDCK1 Siat3	<	40	160	320	80	320	2560	160	640
A/Mauritius/461/2009*	02/07/2009	MDCK1 Siat1	320	320	320	640	320	1280	2560	ND	ND
A/Mauritius/487/2009*	11/07/2009	MDCK1 Siat1	80	80	320	640	80	320	1280	ND	ND
A/Ghana/FS-1031/2009*	13/07/2009	MDCK1 Siat1	40	40	160	80	80	320	640	ND	ND

1. < = <40;

Vaccine strains

* characterised by K158N, N189K, T212A

Table 7. Antigenic analyses of influenza A H3N2 viruses - Guinea Pig RBC - 20nM Oseltamivir

Haemagglutination inhibition titre ¹														
Viruses	Collection Date	Passage History	Post infection ferret sera											
			A/Wis	A/Trieste	A/Bris	A/Uru	A/Fin/	A/Jhb	A/HK	A/HK	A/Perth	A/Wis	A/Niig	A/Niig
			67/05	25c/07	10/07	716/07	9/08	15/08	1952/09	1985/09	16/09	15/09	403/09	403/09
			F1/06	F5/07	F29/08	F26/08	F12/088	F22/08	F22/09	F21/09	F25/09	F24/09	Jap F3/09	Jap F5/09
REFERENCE VIRUSES														
A/Wisconsin/67/2005	31/08/2005	SpfCk3E3 E3	320	320	320	640	160	320		<	<	<	<	<
A/Trieste/25c/2007	Jan-07	MDCKx Siat4	1280	1280	1280	2560	1280	2560	40	80	40	<	40	80
A/Brisbane/10/2007	06/02/2007	E2 E3	1280	1280	1280	2560	1280	1280	<	<	40	<	40	40
A/Uruguay/716/2007	21/06/2007	spfck1, E3 E1	1280	1280	1280	2560	1280	1280	<	<	<	<	<	<
A/Finland/9/2008	07/01/2008	MDCK2 Siat2	640	640	640	1280	640	2560	<	40	40	<	40	40
A/Johannesburg/15/2008	25/06/2008	MDCKx Siat2	320	1280	640	1280	640	2560	<	40	40	<	40	160
A/Hong Kong/1952/2009	24/03/2009	MDCKx2 Siat2	40	40	80	160	80	80	320	1280	2560	640	1280	640
A/Hong Kong/1985/2009	01/04/2009	MDCKx2 Siat2	40	40	80	80	40	80	320	1280	1280	640	2560	640
A/Perth/16/2009	04/07/2009	E3 E1	<	<	<	<	<	<	40	320	640	80	320	160
A/Wisconsin/15/2009	06/07/2009	E2 E1	<	<	<	40	<	<	160	2560	1280	320	640	640
A/Niigata/403/2009		MDCK4 Siat1	80	<	80	80	40	80	320	2560	1280	640	2560	640
TEST VIRUSES														
A/Taiwan/240/2009		E3+2 E1	1280	1280	640	2560	640	640	<	<	<	<	<	<
A/Lithuania/11K/2009	29/04/2009	Siat3	320	1280	640	640	640	640	<	40	<	<	<	40
A/Iceland/6043/2009	30/04/2009	MDCK2 Siat1	320	640	640	1280	640	640	<	40	<	<	<	<
A/Lithuania/142K/2009	16/06/2009	Siat3	640	1280	1280	2560	1280	2560	<	40	80	<	40	160
A/Netherlands/761/2009	26/06/2009	xMdck1 Siat1	80	320	320	640	160	320	<	80	40	<	40	40
A/Ghana/FS-09-1110/2009	27/07/2009	C2 Siat2	160	320	320	640	320	640	40	80	40	<	<	40
A/Cameroon/334/2009	30/07/2009	Siat3	160	320	320	640	320	640	40	80	80	40	80	80
A/Cameroon/350/2009	05/08/2009	Siat3	160	320	320	640	160	640	40	80	80	40	80	80
A/Iran/755/2009	Jun-09	MDCK2 Siat2	40	40	80	160	40	80	640	2560	5120	1280	1280	1280
A/Columbia/227/2009	01/05/2009	C4 Siat1	<	<	80	160	40	40	160	1280	640	320	640	320
A/Luxembourg/791/2009	11/05/2009	MDCK3 Siat3	40	40	80	160	40	80	320	5120	2560	640	2560	640
A/Mauritius/380/2009	05/06/2009	C1 Siat1	80	80	320	640	160	160	320	1280	1280	640	1280	640
A/Mauritius/511/2009	17/07/2009	C1 Siat1	160	160	320	320	160	160	320	1280	1280	640	1280	640
A/Lithuania/131K/2009	23/07/2009	Siat3	40	40	80	160	40	40	320	2560	1280	640	1280	640
A/Ghana/FS-09-1212/2009	10/08/2009	C2 Siat2	<	40	80	80	40	40	320	1280	2560	640	1280	640
A/Ghana/FS-09-1267/2009	13/08/2009	C2 Siat2	<	<	80	80	<	40	320	2560	5120	1280	1280	1280

1. < = <40

Vaccine strains

Table 8. Antigenic analysis of influenza A H3N2 viruses - Plaque Reduction Neutralization - MDCK-SIAT

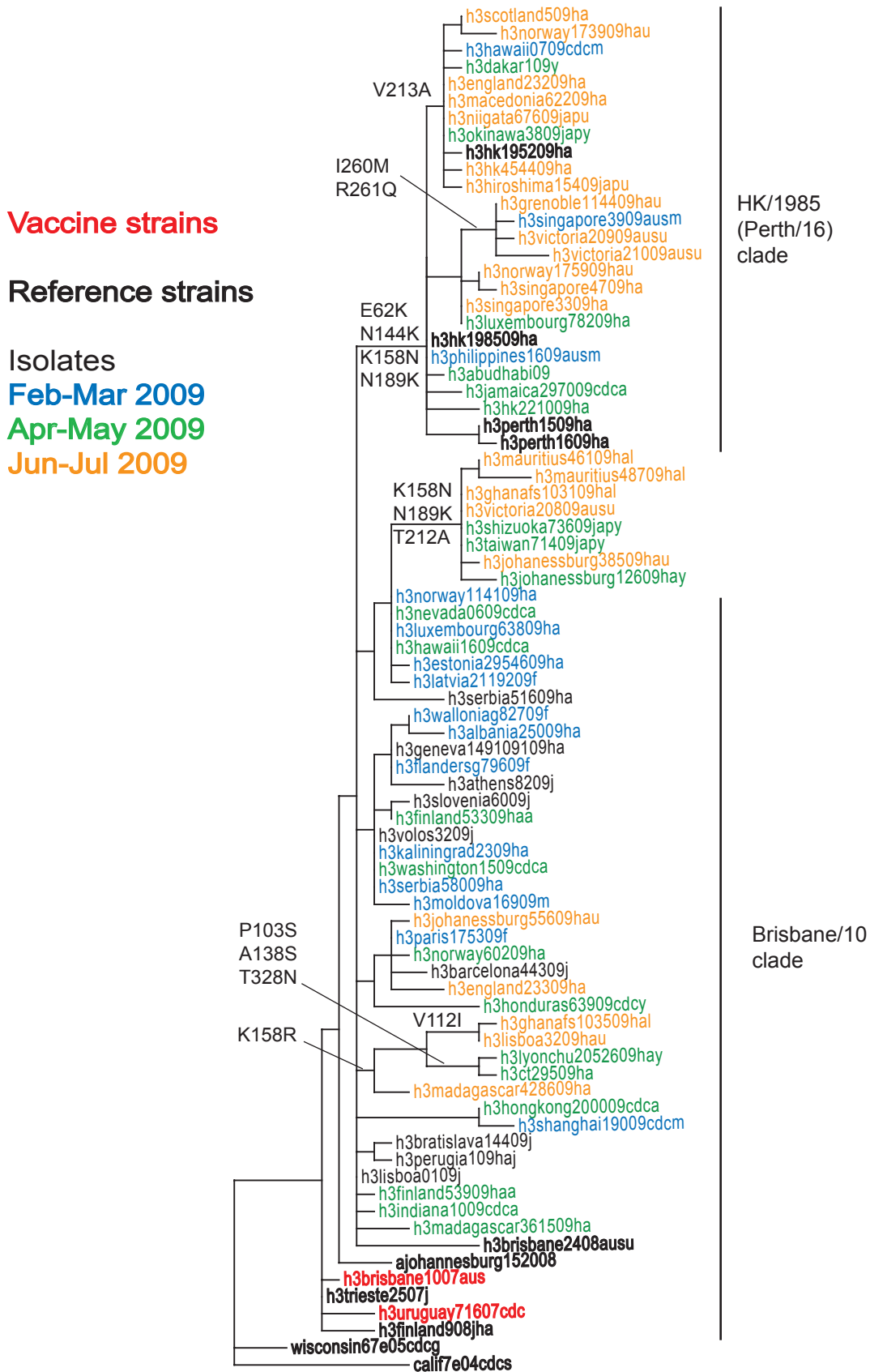
Viruses	Collection Date	Passage History	Virus Neutralization titre ¹				
			Post infection ferret sera ²				
			A/Well 1/04 F13/04	A/Bris 10/07 F18/07	A/Uru 716/07 F3/07	A/Eng 394/08 F32/08	A/Bris 24/08 F3/09
REFERENCE VIRUSES							
A/Wellington/1/2004	26/01/2004	E3 \1	5120	640	640	320	1280
A/Brisbane/10/2007	06/02/2007	E2 \3	160	2560	5120	320	2560
A/Uruguay/716/2007	21/06/2007	spfck1, E3 \1	160	2560	5120	320	2560
A/England/394/2008	22/09/2008	TC P2 \1	40	1280	1280	1280	1280
A/Brisbane/24/08	23/06/2008	E5	80	2560	2560	160	2560
TEST VIRUSES							
A/Thessaloniki/16/2009	15/01/2009	p3 \1	40	1280	1280	320	1280
A/St. Petersburg/5/2009	16/02/2009	C2 \1	80	1280	1280	320	1280
A/Aragon/RR-3438/2009	29/12/2008	MDCK1 \1	320	1280	1280	320	1280
A/Geneva/1570509/2009	14/02/2009	\1	40	1280	2560	640	2560
A/Geneva/1514818/2009	30/01/2009	\1	40	2560	5120	640	2560
A/Hong Kong/1926/2009	23/03/2009	MDCKx2 \1	<	160	80	160	160
A/Hong Kong/1985/2009	01/04/2009	MDCKx2 \1	<	160	80	160	160

¹ Based on 50% plaque reduction compared to serum negative controls

² RDE treated

Vaccine strain

Figure 8. Phylogenetic comparison of H3N2 HA genes



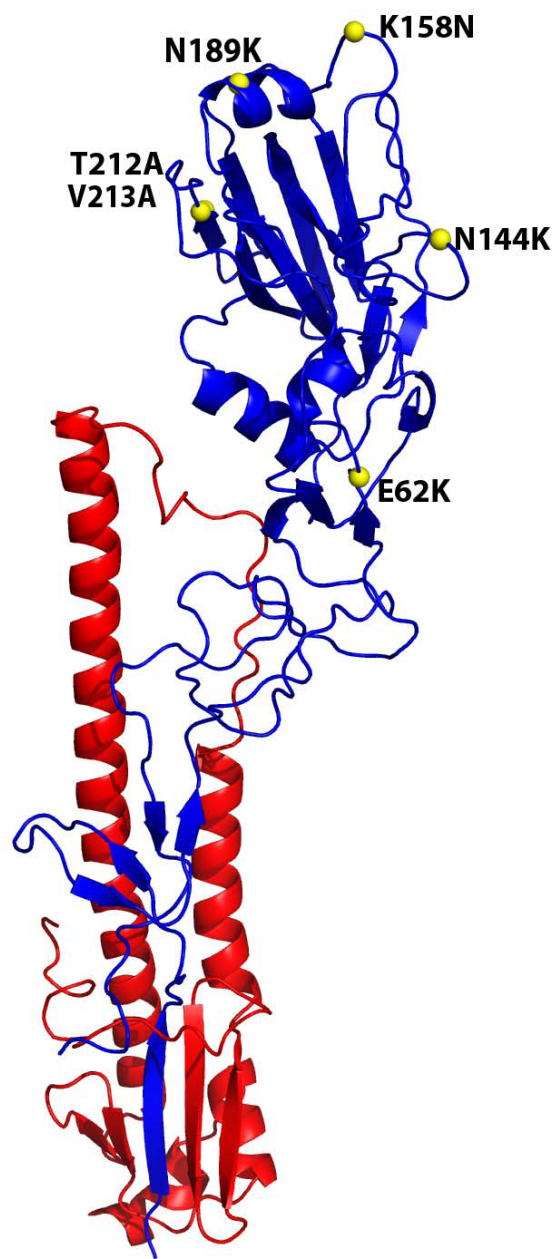


Figure 9. Locations of amino acid changes characteristic of A/Perth/16/2009-like viruses on the H3 HA structure

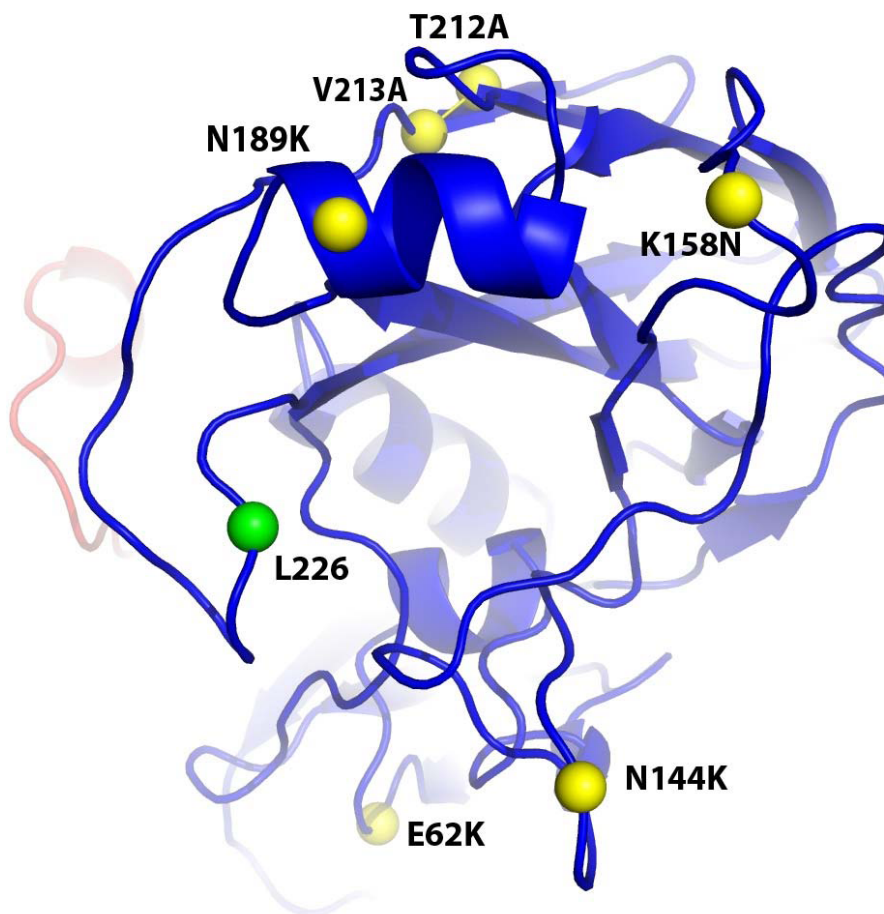


Figure 10. Phylogenetic comparison of H3N2 NA genes

Vaccine strains

Reference strains

Isolates

Feb-Mar 2009

Apr-May 2009

Jun-Jul 2009

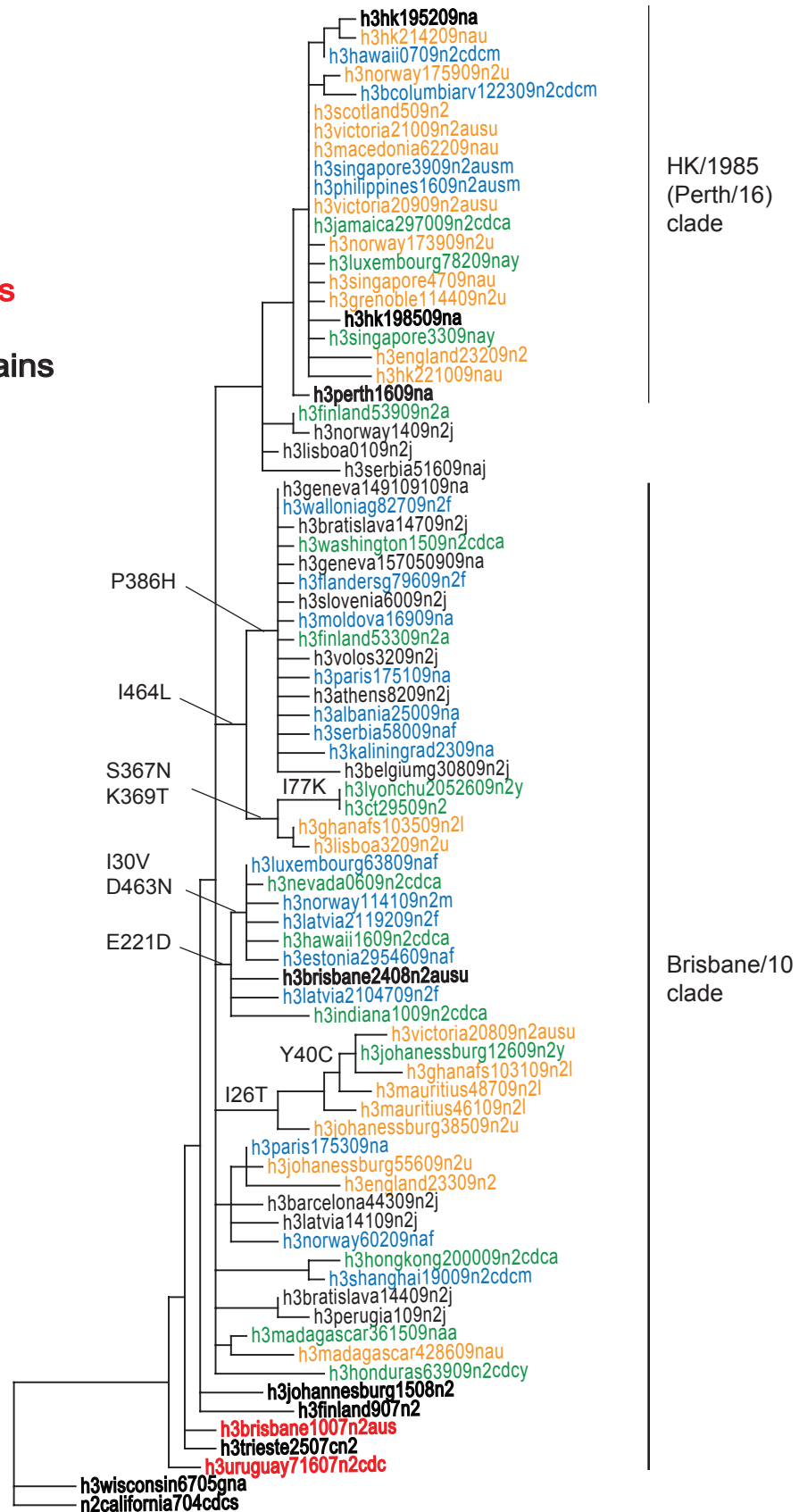


Table 9. Antigenic analyses of influenza B viruses (Victoria lineage)

Haemagglutination inhibition titre ¹												
Viruses	Collection date	Passage History	Post infection ferret sera									
			B/Shan ²	B/Shan	B/Bris	B/HK	B/Mal	B/Vic	B/Eng	B/Bris	B/Bris	B/Paris
			7/97	7/97 F12/02	32/02 F2/03	45/05 F15/05	2506/04 F28/05	304/06 F16/06	393/08 F31/08	33/08 F1/09	60/08 F2/09	1762/08 F11/09
REFERENCE VIRUSES												
B/Shandong/7/97		Ex	160	40	160	<	160	320	320	320	640	<
B/Brisbane/32/2002	02/07/2002	E2 \6	320	20	40	40	80	40	20	<	20	<
B/HK/45/2005	07/02/2005	MDCK2 Siat4	640	80	160	160	80	40	40	<	<	<
B/Malaysia/2506/2004	06/12/2004	E3 \3	320	80	80	<	160	160	160	80	160	<
B/Victoria/304/2006	06/06/2006	E2 \2	320	40	80	80	160	160	80	40	160	20
B/England/393/2008	29/08/2008	E1 \2	160	40	40	<	80	160	160	160	320	<
B/Brisbane/33/2008	13/07/2008	E3 \1	160	40	160	<	160	320	320	160	640	20
B/Brisbane/60/2008	04/08/2008	E4 \1	640	80	160	10	320	160	80	40	160	20
B/Paris/1762/2008	09/02/2009	C1 Siat2	320	<	<	<	<	20	40	10	20	40
TEST VIRUSES												
B/Iwate/17/2009		MDCK2+2 Siat1	640	<	20	160	<	10	<	<	<	<
B/Hubei-xiling/37/2009		E3+2 E1	640	40	80	80	160	80	40	20	40	<
B/Sabac/1060/2009	03/03/2009	Primo Siat1	320	10	10	160	40	20	40	<	<	ND
B/Belgrade/1094/2009	03/03/2009	MDCK1 Siat1	640	10	10	160	40	10	20	<	20	ND
B/Hong Kong/128/2009	06/04/2009	MDCKx2 Siat1	320	10	10	160	20	10	10	<	20	ND
B/Singapore/21/2009	02/05/2009	MDCK 1 Siat1	640	<	40	160	80	20	20	10	<	<
B/Hong Kong/235/2009	09/07/2009	MDCKx2 Siat1	320	<	20	80	<	<	<	<	<	<
B/Okinawa/10/2009		MDCK1+2 Siat1	320	<	<	20	<	<	40	20	40	80
B/Sabac/1232/2009	11/03/2009	MDCK1 Siat1	640	10	<	10	20	20	40	<	40	80
B/St. Petersburg/103/2009	25/03/2009	C1 Siat1	320	10	20	<	<	10	10	<	40	80
B/Georgia/1307 (39G)/2009	02/03/2009	CII Siat1	320	<	20	<	20	20	80	20	40	80
B/Georgia/1310 (41G)/2009	16/03/2009	CII Siat1	320	<	10	<	10	20	40	20	40	ND
B/Ukraine/104/09	05/03/2009	MDCK1 Siat2	320	<	10	<	10	10	20	<	40	ND
B/Geneva/1724714/2009	31/03/2009	x Siat4	320	<	<	<	10	<	<	<	80	ND
B/St. Petersburg/61/2009	16/03/2009	E1 E1	160	10	80	<	80	80	160	320	320	ND
B/Zagreb/4596/2009	20/03/2009	E2 E1	320	10	160	<	160	80	160	320	640	ND
B/Zagreb/46142009	20/03/2009	E2 E1	160	10	80	<	80	40	80	80	160	ND
B/Iceland/4744/2009	02/04/2009	MDCK1 Siat1	1280	<	20	80	10	40	160	80	160	160
B/Netherlands/751/2009	07/04/2009	Mk1 Siat1	320	<	<	20	<	<	40	20	40	ND
B/Kiev/195/09	15/04/2009	MDCK1 Siat2	1280	80	40	20	80	40	320	80	320	ND
B/Finland/542/2009	16/04/2009	MDCK1 Siat1	320	20	10	<	40	20	160	40	160	ND
B/Madagascar/3747/2009	19/04/2009	MDCK1 Siat1	320	<	10	10	40	20	40	20	80	40
B/St. Petersburg/52/2009	03/04/2009	E2 Siat1	320	20	160	<	160	80	80	160	320	ND
B/Norway/1854/2009	04/05/2009	MDCK1 Siat1	320	<	<	20	<	<	40	40	80	20
B/Norway/1847/2009	08/05/2009	MDCK1 Siat1	320	<	<	<	<	<	10	10	20	40
B/Norway/1474/2009	18/05/2009	MDCK1 Siat1	320	<	<	<	20	<	40	<	160	ND
B/Madagascar/4254/2009	02/06/2009	MDCK1 Siat1	160	<	<	10	20	<	<	<	80	ND
B/Iceland/8611/2009	19/06/2009	MDCK1 Siat1	320	<	<	<	<	<	20	10	20	40
B/Ghana/FS-1038/2009	13/07/2009	MDCK1 Siat2	640	<	40	20	40	40	160	40	80	160
B/Mauritius/515/2009	18/07/2009	MDCK1 Siat1	320	<	<	<	<	<	10	10	20	40

1. <= <10; 2. hyperimmune sheep serum

Figure 11. Phylogenetic comparison of influenza B HA genes (Victoria lineage)

Vaccine strain

Reference strains

Isolates

Jan - Feb 2009

Mar - Apr 2009

May - Jul 2009

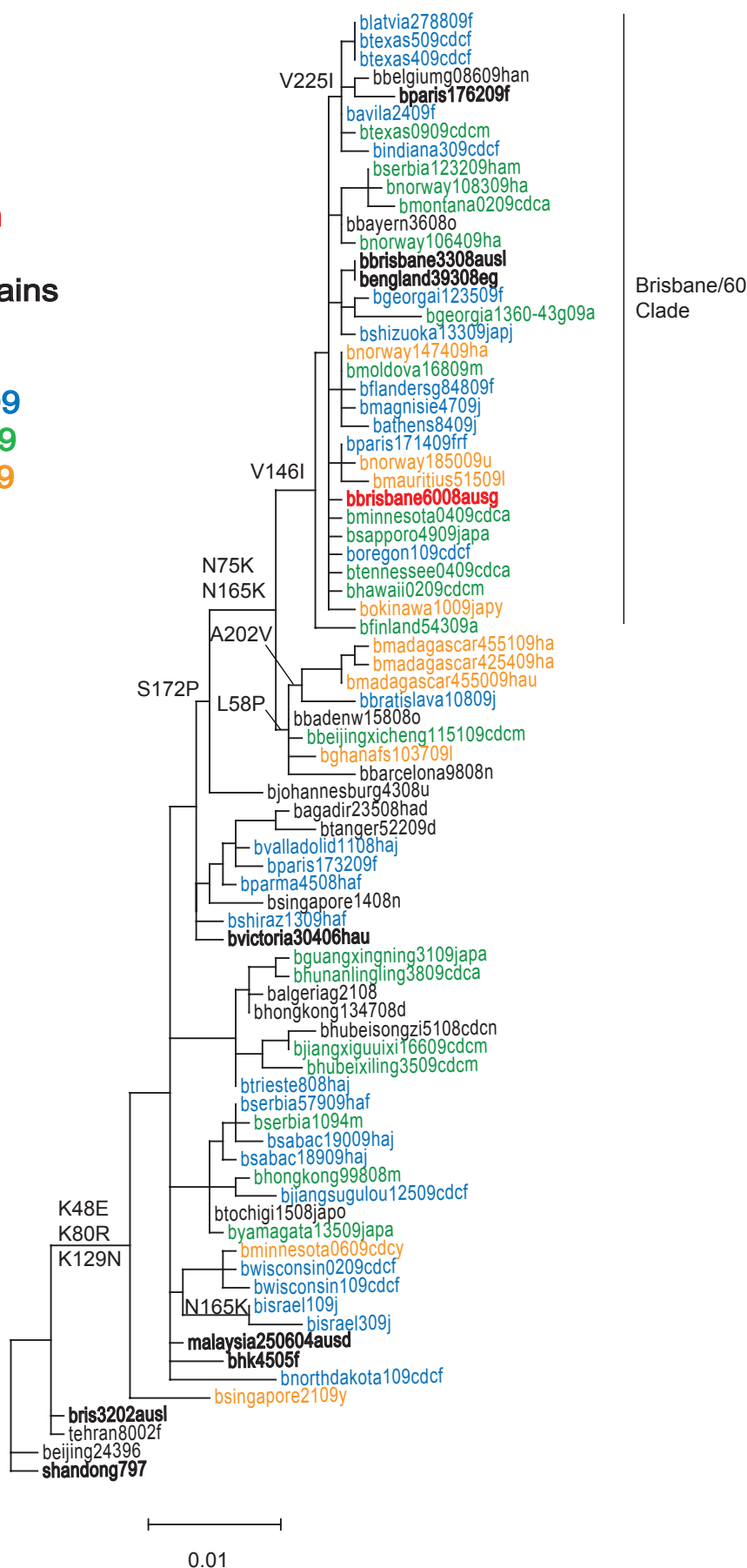


Figure 12. Phylogenetic comparison of influenza B NA genes

Vaccine strain

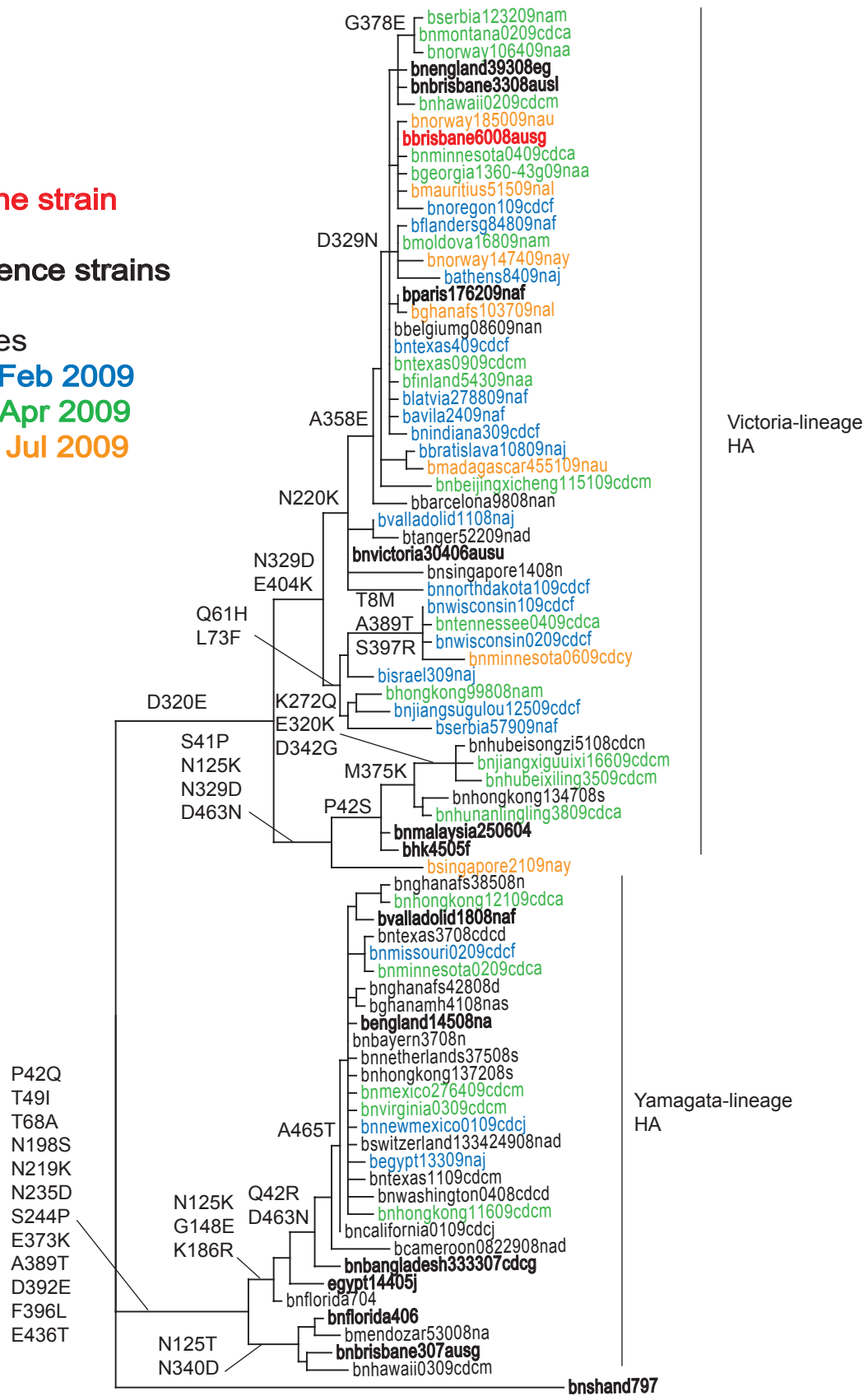
Reference strains

Isolates

Jan - Feb 2009

Mar - Apr 2009

May - Jul 2009



0.01

Table 10. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹						
			B/FI ²	Post infection ferret sera					
				B/Eg	B/FI	B/Bris	B/Eng	B/Vall	B/Ban
				4/06 Sh 478	144/05 F3/07	4/06 F21/07	3/07 F24/07	145/08 F9/08	18/08 F10/08
REFERENCE VIRUSES									
B/Egypt/144/2005	01/05/2005	E2 E2	1280	80	160	320	40	10	80
B/Florida/4/2006	15/12/2006	E2 E2	5120	160	640	640	80	20	160
B/Brisbane/3/2007	03/09/2007	E2 E2	1280	80	320	320	40	10	80
B/England/145/2008		Ex E3	160	<	<	20	40	<	10
B/Valladolid/18/2008	18/02/2008	MDCK1 Siat4	2560	80	80	80	160	160	160
B/Bangladesh/3333/2007	19/08/2007	E4 E2	640	40	40	80	20	10	80
TEST VIRUSES									
B/Latvia/2-30/2009	02/02/2009	MDCK Siat1	5120	160	160	160	320	320	160
B/Mashad/6/2009	08/02/2009	2 Siat1	2560	80	80	160	80	160	160
B/Zahedan/7/2009	12/02/2009	2 Siat1	1280	80	80	160	80	80	80
B/Aswan/1895/2009	15/02/2009	Cx Siat1	2560	160	80	80	80	160	160
B/Hong Kong/105/2009	20/03/2009	MDCKx2 Siat1	2560	80	160	160	160	160	160
B/Alexandria/1133/2009	20/03/2009	Cx Siat1	5120	160	160	160	160	320	320
B/Alexandria/1142/2009	22/03/2009	Cx Siat1	1280	80	40	40	80	80	80
B/Aswan/1978/2009	07/04/2009	Cx Siat1	2560	80	80	40	80	160	160
B/Alexandria/1139/2009	10/04/2009	Cx Siat1	2560	40	40	80	80	160	80
B/Aswan/1983/2009	11/04/2009	Cx Siat1	2560	80	80	80	80	160	160
B/Tehran/8/2009	16/05/2009	2 Siat1	1280	80	40	160	80	80	80
B/Norway/1440/2009	20/05/2009	MDCK1 Siat1	2560	80	80	160	80	160	160
B/Stockholm/4/2009	20/05/2009	1 Siat1	5120	160	320	640	320	320	320
B/Hong Kong/251/2009	13/07/2009	MDCKx2 Siat1	2560	80	80	80	80	160	160

1. < = <10; 2. hyperimmune sheep serum

Figure 13. Phylogenetic comparison of influenza B HA genes (Yamagata lineage)

Reference strains

Isolates

Dec 2008 - Jan 2009

Feb - Mar 2009

Apr - May 2009

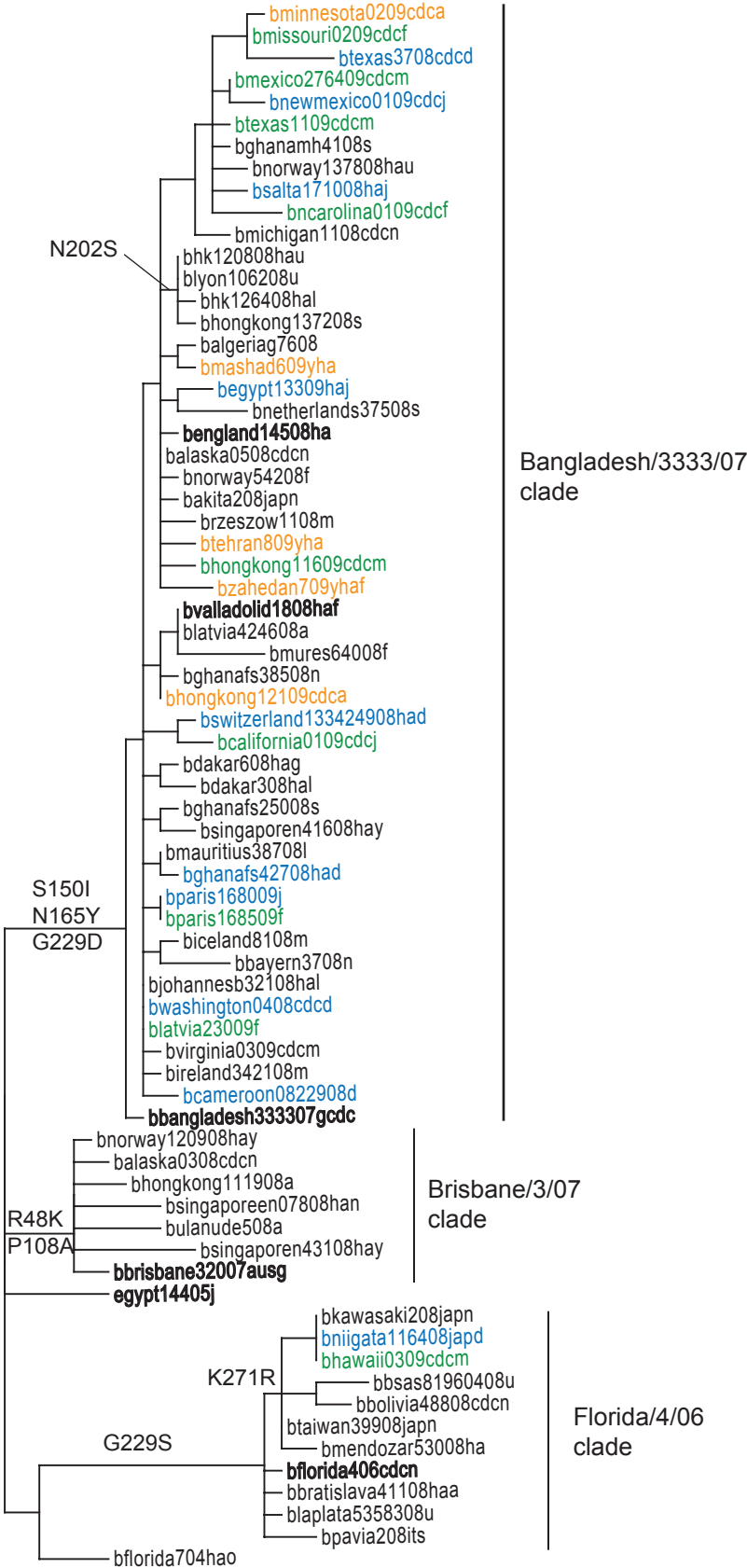


Figure 14. Phylogenetic comparison of influenza H5 HA genes

**Candidate
Vaccine strains**

2009 viruses

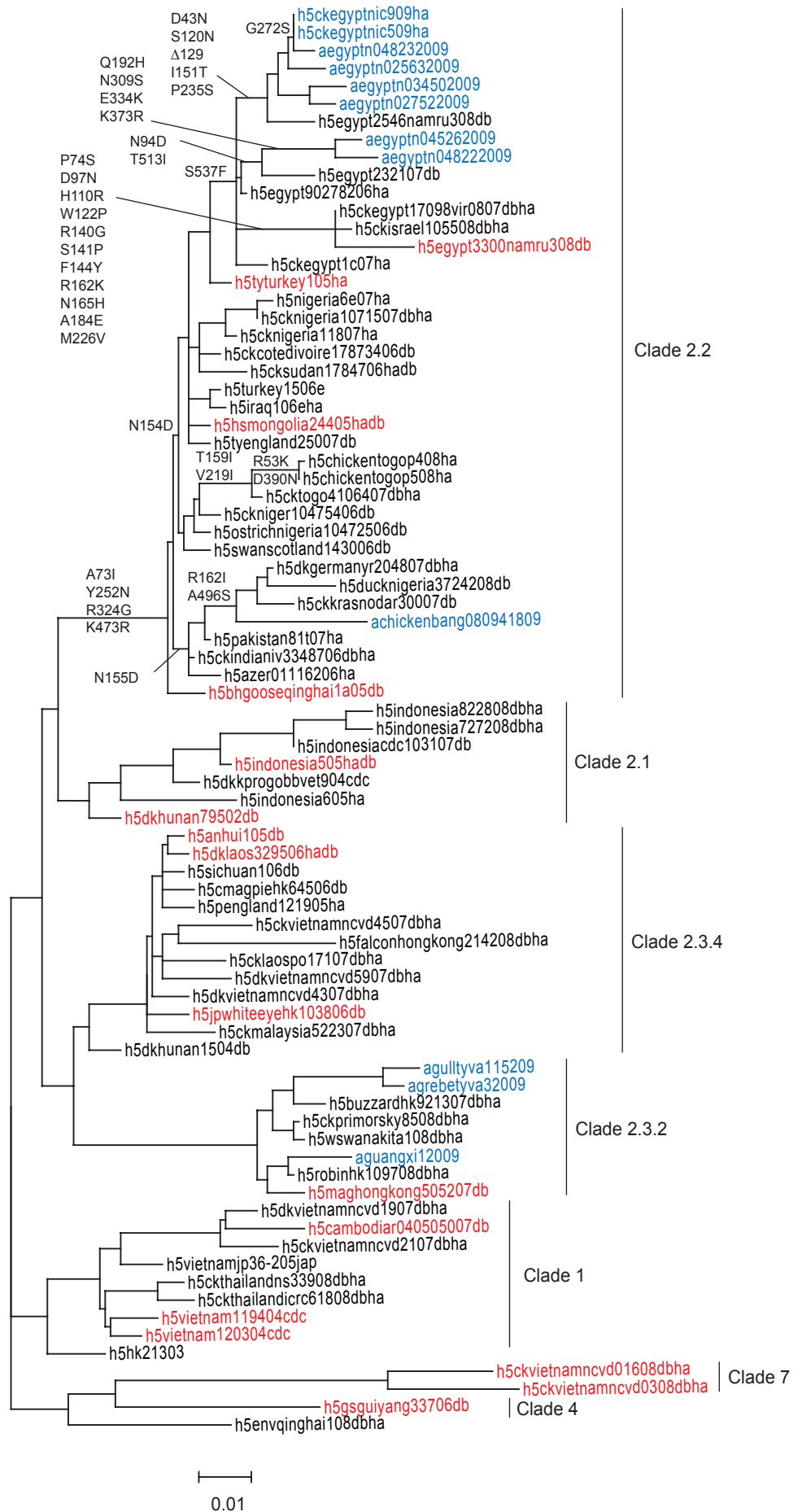


Figure 15. Phylogenetic comparison of influenza H5 NA genes

